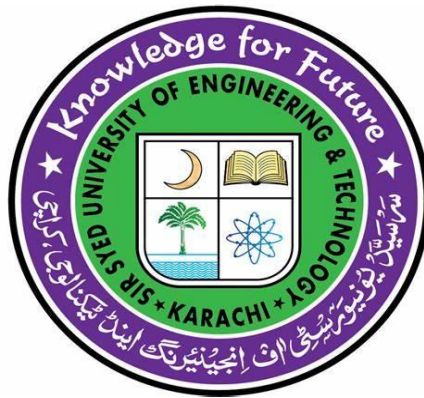


Laboratory Manual

Object Oriented Programming

(CS – 127L)

2nd Semester Spring 2023



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Lab # 1

Task 1: Open Visual Studio and run your Console Application with
- First Name - Last Name - Contact No - Email Address - Address

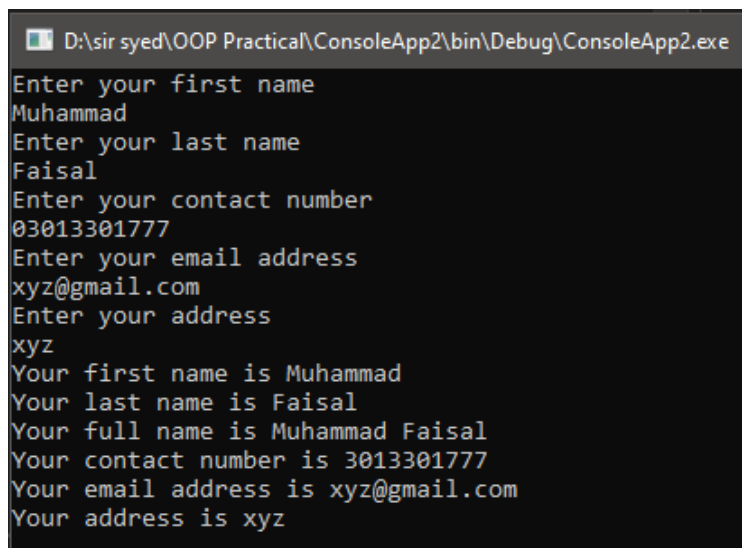
Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    internal class Program
    {
        public string n1, n2,e,a;
        public long num;
        public void details()
        {
            Console.WriteLine("Enter your first name");
            n1= Console.ReadLine();
            Console.WriteLine("Enter your last name");
            n2= Console.ReadLine();
            Console.WriteLine("Enter your contact number");
            num = Convert.ToInt64(Console.ReadLine());
            Console.WriteLine("Enter your email address");
            e= Console.ReadLine();
            Console.WriteLine("Enter your address");
            a= Console.ReadLine();
        }

        public void display()
        {
            Console.WriteLine("Your first name is " + n1);
            Console.WriteLine("Your last name is " + n2);
            Console.WriteLine("Your full name is " + n1 + " " + n2);
            Console.WriteLine("Your contact number is " + num);
            Console.WriteLine("Your email address is " + e);
            Console.WriteLine("Your address is " + a);
        }
        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.details();
            obj.display();
            Console.ReadKey();
        }
    }
}
```

Output



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Enter your first name
Muhammad
Enter your last name
Faisal
Enter your contact number
03013301777
Enter your email address
xyz@gmail.com
Enter your address
xyz
Your first name is Muhammad
Your last name is Faisal
Your full name is Muhammad Faisal
Your contact number is 3013301777
Your email address is xyz@gmail.com
Your address is xyz
```

Task 2: Let us consider a Rectangle object. It has attributes such as length and width. Depending upon the design, it may need ways for accepting the values of these attributes, calculating the area, and displaying details.

Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

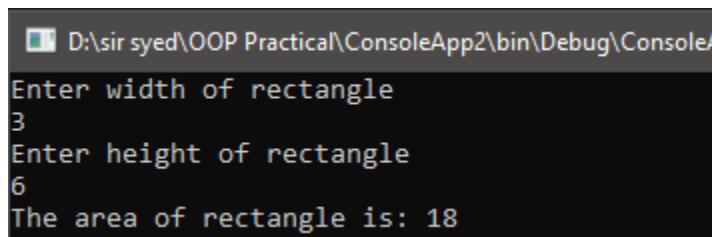
namespace ConsoleApp2
{
    internal class Program
    {
        public int n1,n2,a;

        public void input()
        {
            Console.WriteLine("Enter width of rectangle");
            n1=Convert.ToInt32(Console.ReadLine());

            Console.WriteLine("Enter height of rectangle");
            n2 = Convert.ToInt32(Console.ReadLine());
        }
        public void area()
        {
            a = n1 * n2;
        }

        public void display()
        {
            Console.WriteLine("The area of rectangle is: "+a);
        }
        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.input();
            obj.area();
            obj.display();
            Console.ReadLine();
        }
    }
}
```

Output

A screenshot of a Windows Command Prompt window. The title bar shows the file path: D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe. The prompt displays the following text: "Enter width of rectangle", followed by the user input "3", then "Enter height of rectangle", followed by the user input "6", and finally the output "The area of rectangle is: 18".

```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Enter width of rectangle
3
Enter height of rectangle
6
The area of rectangle is: 18
```

Task 3: Write an application that asks the user to enter two numbers, obtains the two numbers from the user and prints the sum, product, difference and quotient of the two numbers.

Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
```

```
namespace ConsoleApp2
```

```
{
```

```

class Program
{
    public int n1, n2, sum, product, diff, quotient;

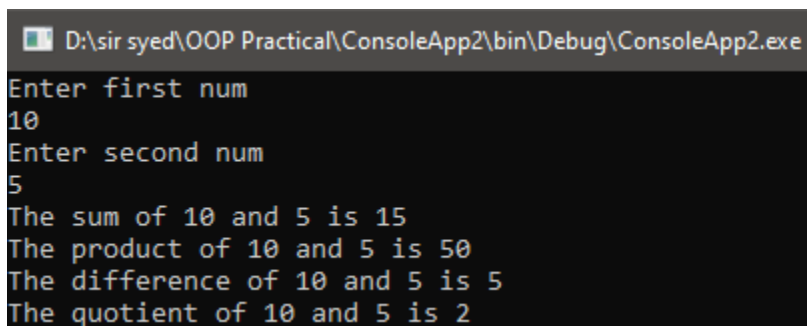
    public void input()
    {
        Console.WriteLine("Enter first num");
        n1 = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Enter second num");
        n2 = Convert.ToInt32(Console.ReadLine());
    }
    public void add()
    {
        sum = n1 + n2;
    }
    public void sub()
    {
        diff = n1 - n2;
    }
    public void multiply()
    {
        product = n1 * n2;
    }
    public void divide()
    {
        quotient = n1 / n2;
    }
    public void display()
    {
        Console.WriteLine("The sum of {0} and {1} is {2} ", n1,n2,sum);
        Console.WriteLine("The product of {0} and {1} is {2} ", n1, n2, product);
        Console.WriteLine("The difference of {0} and {1} is {2} ", n1, n2, diff);
        Console.WriteLine("The quotient of {0} and {1} is {2} ", n1, n2, quotient);
    }
}

class Program2 {
static void Main(string[] args)
{
    Program obj = new Program();
    obj.input();
    obj.add();
    obj.sub();
    obj.multiply();
    obj.divide();
    obj.display();
    Console.ReadKey();
}
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Enter first num
10
Enter second num
5
The sum of 10 and 5 is 15
The product of 10 and 5 is 50
The difference of 10 and 5 is 5
The quotient of 10 and 5 is 2

```

Task 4: Write an application that inputs from the user the radius of a circle and prints the circle's diameter, circumference and area. Use the following formulas (r is the radius): diameter = 2r, circumference = $2\pi r$, area = πr^2 .

Code

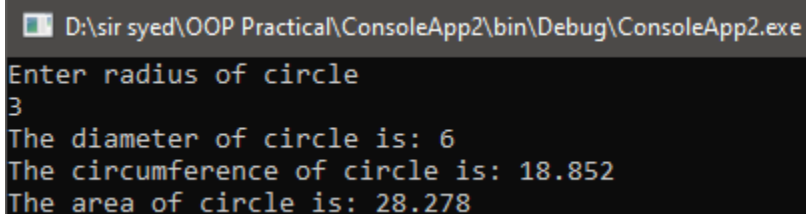
```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    class Program
    {
        public double n1, c, a, d;

        public void input()
        {
            Console.WriteLine("Enter radius of circle");
            n1 = Convert.ToDouble(Console.ReadLine());
        }
        public void diameter()
        {
            d = n1 + n1;
        }
        public void circumference()
        {
            c = 2*3.142*n1;
        }
        public void area()
        {
            a = 3.142*(n1 *n1);
        }
        public void display()
        {
            Console.WriteLine("The diameter of circle is: "+d);
            Console.WriteLine("The circumference of circle is: "+ c);
            Console.WriteLine("The area of circle is: "+a);
        }
    }

    class Program2 {
        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.input();
            obj.diameter();
            obj.circumference();
            obj.area();
            obj.display();
            Console.ReadLine();
        }
    }
}
```

Output



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Enter radius of circle
3
The diameter of circle is: 6
The circumference of circle is: 18.852
The area of circle is: 28.278
```

Lab # 2

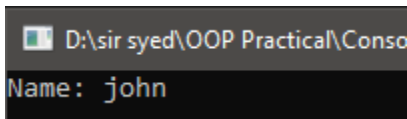
Task 1: Create a variable called name of type string and assign it the value "John"

Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    class Program
    {
        public string name="john";
        public void display()
        {
            Console.WriteLine("Name: "+name);
        }
    }
    class Program2 {
        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.display();
            Console.ReadLine();
        }
    }
}
```

Output

A screenshot of a Windows command prompt window. The title bar shows the path 'D:\sir syed\OOP Practical\ConsoleApp2'. The command prompt displays the output 'Name: john'.

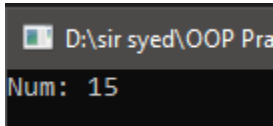
Task 2: Create a variable called myNum of type int and assign it the value 15

Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    class Program
    {
        public int num=15;
        public void display()
        {
            Console.WriteLine("Name: "+num);
        }
    }
    class Program2 {
        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.display();
            Console.ReadLine();
        }
    }
}
```

Output



D:\sir syed\OOP Pra
Num: 15

Task 3: Declare a variable without assigning the value, and assign the value later:

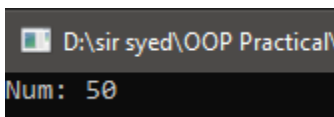
Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    class Program
    {
        public int num;
        public void display()
        {
            Console.WriteLine("Num: "+num);
        }
    }

    class Program2 {
        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.num = 50;
            obj.display();
            Console.ReadLine();
        }
    }
}
```

Output



D:\sir syed\OOP Practical\N
Num: 50

Task 4: Perform Decision Making: Equality and Relational Operators program given in theory lecture using Classes, Variable and Objects.

Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    class Program
    {
        public int num;
        public void input()
        {
            Console.WriteLine("Enter temprature");
            num=Convert.ToInt32(Console.ReadLine());
        }
        public void Decision()
        {
            if (num == 100)
            {

```

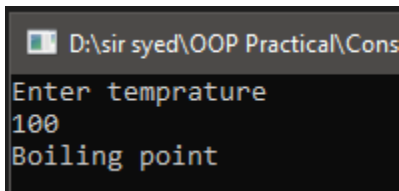
```

        Console.WriteLine("Boiling point");
    }
    else if (num >= 25)
    {
        Console.WriteLine("Very hot");
    }
    else if(num >=15 && num<=25)
    {
        Console.WriteLine("Warm");
    }
    else
    {
        Console.WriteLine("cold");
    }
}
}

class Program2 {
static void Main(string[] args)
{
    Program obj = new Program();
    obj.input();
    obj.Decision();
    Console.ReadLine();
}
}
}

```

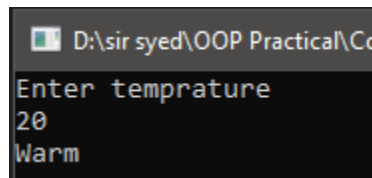
Output



```

D:\sir syed\OOP Practical\Cons
Enter temprature
100
Boiling point

```



```

D:\sir syed\OOP Practical\Co
Enter temprature
20
Warm

```

Task 5: Make a class EvenODD, which checks whether the number you inputted is an Even or Odd number.

Code

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    class Evenodd
    {
        public int num;
        public void input()
        {
            Console.WriteLine("Enter number");
            num=Convert.ToInt32(Console.ReadLine());
        }
        public void Decision()
        {
            if (num%2 ==1)
            {
                Console.WriteLine("{0} is an odd number", num);
            }
            else
            {
                Console.WriteLine("{0} is an even number", num);
            }
        }
    }

    class Program2 {
static void Main(string[] args)

```

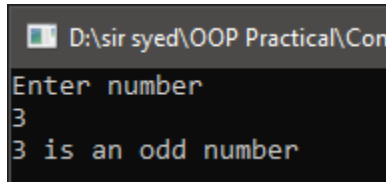


```

    {
        Evenodd obj = new Evenodd();
        obj.input();
        obj.Decision();
        Console.ReadLine();
    }
}

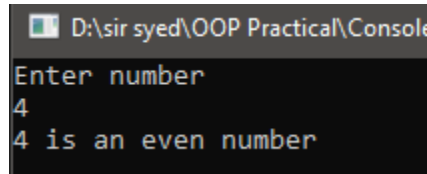
```

Output



D:\sir syed\OOP Practical\Console

Enter number
3
3 is an odd number



D:\sir syed\OOP Practical\Console

Enter number
4
4 is an even number

Task 6: Give real time examples (3) of four principles of OOP in your own words

1. Encapsulation: Encapsulation is the practice of hiding the internal details of an object and providing a public interface to interact with it. A good example of encapsulation is a smartphone. The smartphone has a lot of hardware components and software processes that work together to provide various functionalities like calling, texting, browsing the internet, taking pictures, etc. However, as a user, we only interact with the user interface, which is a public interface to access the functionality provided by the smartphone. We don't need to know how the software or hardware components work internally.
2. Encapsulation: Encapsulation is the practice of hiding the internal details of an object and providing a public interface to interact with it. A good example of encapsulation is a smartphone. The smartphone has a lot of hardware components and software processes that work together to provide various functionalities like calling, texting, browsing the internet, taking pictures, etc. However, as a user, we only interact with the user interface, which is a public interface to access the functionality provided by the smartphone. We don't need to know how the software or hardware components work internally.
3. Inheritance: Inheritance is a way to create a new class by inheriting the properties and methods of an existing class. A real-time example of inheritance is a zoo. In a zoo, there are different animals with common characteristics like eating, sleeping, and reproducing. So, instead of creating separate classes for each animal, a base class called "Animal" can be created that contains these common characteristics. Then, child classes like "Lion," "Tiger," and "Elephant" can be created that inherit from the "Animal" class and add their own unique characteristics.
4. Polymorphism: Polymorphism is the ability of an object to take on many forms. A good example of polymorphism is a "Shape" class. A "Shape" class can have different child classes like "Circle," "Rectangle," "Triangle," etc., that can have their own implementation of the "Area" method. The "Area" method can take different forms depending on the child class, but they all provide an implementation of the "Area" method, which makes it polymorphic.
5. Abstraction: Abstraction is the practice of hiding unnecessary details and showing only the relevant information to the user. A real-time example of abstraction is a car. As a user, we don't need to know how the engine works or how the fuel is burned to move the car. All we need to know is how to operate the steering wheel, brake, accelerator, and gear lever. The car provides a simple interface to interact with it, hiding all the unnecessary details.

Lab # 3

Task 1: Write the output of the following program.

```
using System;
public class Book
{
    public string Title;
    public string Author;
    public short YearPublished;
    public int NumberOfPages;
    public char CoverType;
}

public class Exercise
{

    static void Main()
    {
        var First = new Book();

        First.Title = " Turbo C ";
        First.Author = "Robert Lafore";
        First.YearPublished = 1996;
        First.NumberOfPages = 872;
        First.CoverType = 'H';
        Console.WriteLine("Book Characteristics");

        Console.Write("Title: ");
        Console.WriteLine(First.Title);
        Console.Write("Author: ");
        Console.WriteLine(First.Author);
        Console.Write("Year:  ");
        Console.WriteLine(First.YearPublished);
        Console.Write("Pages: ");
        Console.WriteLine(First.NumberOfPages);
        Console.Write("Cover: ");
        Console.WriteLine(First.CoverType);
        Console.ReadKey();
    }
}
```

Output

```
Book Characteristics
Title: Turbo C
Author: Robert Lafore
Year: 1996
Pages: 872
Cover: H
```

Task 2: Create a class student with a data members name, age, marks of English , marks of math, marks of science, total marks, obtained marks and percentage provide member functions CalculateTotalMarks and CalculatePercentage to calculate marks and percentage in main.

Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    internal class Student
    {
        public string name;
```

```

public int age;
public double marks_Eng, marks_math, marks_sci, total_marks = 300, obtained_marks, per;
public void details()
{
    Console.WriteLine("Enter name of student");
    name = Console.ReadLine();

    Console.WriteLine("Enter age of student");
    age = Convert.ToInt32(Console.ReadLine());

    Console.WriteLine("Enter marks of math");
    marks_math = Convert.ToDouble(Console.ReadLine());

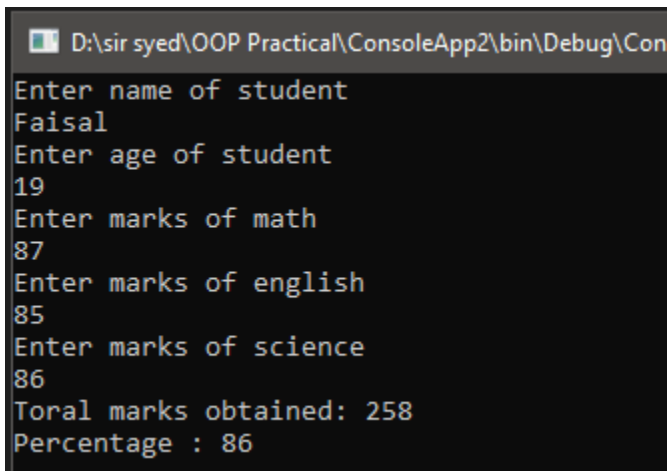
    Console.WriteLine("Enter marks of english");
    marks_Eng = Convert.ToDouble(Console.ReadLine());

    Console.WriteLine("Enter marks of science");
    marks_sci = Convert.ToDouble(Console.ReadLine());
}
public void CalculateTotalMarks()
{
    obtained_marks = marks_math + marks_Eng + marks_sci;
    Console.WriteLine("Total marks obtained: " + obtained_marks);
}

public void percentage()
{
    per = (obtained_marks / total_marks) * 100;
    Console.WriteLine("Percentage : " + per);
}
static void Main(string[] args)
{
    Student obj = new Student();
    obj.details();
    obj.CalculateTotalMarks();
    obj.percentage();
    Console.ReadLine();
}
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Enter name of student
Faisal
Enter age of student
19
Enter marks of math
87
Enter marks of english
85
Enter marks of science
86
Total marks obtained: 258
Percentage : 86

```

Task 3: Write a program that contains a class which has a method that takes user name as input and second functions which returns number of vowels present in it and Main program prints the number of vowels.

Code

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2

```

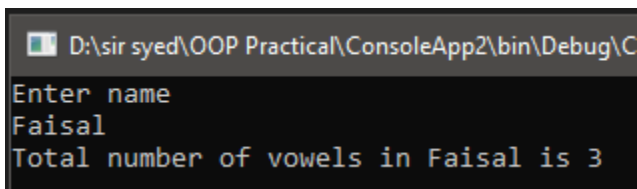
```

{
    internal class Program
    {
        public string name;
        public int count = 0;
        public void Username()
        {
            Console.WriteLine("Enter name");
            name = Console.ReadLine();
        }
        public int Vowels()
        {
            for(int i=0; i<name.Length; i++)
            {
                if (name[i]=='a' || name[i] == 'A')
                {
                    count++;
                }
                else if (name[i] == 'e' || name[i] == 'E')
                {
                    count++;
                }
                else if (name[i] == 'i' || name[i] == 'I')
                {
                    count++;
                }
                else if (name[i] == 'o' || name[i] == 'O')
                {
                    count++;
                }
                else if (name[i] == 'u' || name[i] == 'U')
                {
                    count++;
                }
            }
            return count;
        }

        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.Username();
            Console.WriteLine("Total number of vowels in {0} is {1} ",obj.name,obj.Vowels());
            Console.ReadLine();
        }
    }
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\C
Enter name
Faisal
Total number of vowels in Faisal is 3

```

Task 4: Write a program that performs typecasting of atleast 3 predefined datatypes in C#.

Code

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    class Program
    {

```

```

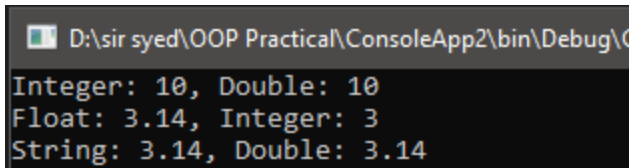
static void Main(string[] args)
{
    int numInt = 10;
    double numDouble = (double)numInt;
    Console.WriteLine("Integer: {0}, Double: {1}", numInt, numDouble);

    float numFloat = 3.14f;
    int numInt2 = (int)numFloat;
    Console.WriteLine("Float: {0}, Integer: {1}", numFloat, numInt2);

    string strDouble = "3.14";
    double numDouble2 = Convert.ToDouble(strDouble);
    Console.WriteLine("String: {0}, Double: {1}", strDouble, numDouble2);
    Console.ReadLine();
}
}
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\O
Integer: 10, Double: 10
Float: 3.14, Integer: 3
String: 3.14, Double: 3.14

```

Task 5: Write program that checks whether the given number is prime or not. (Prime number logic: a number is prime if it is divisible only by one and itself. Remember two is the only even and also the smallest prime number. First few prime numbers are 2, 3, 5, 7, 11, 13, 17.... etc.).

Code

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    internal class Program
    {
        public int num;
        public void input()
        {
            Console.WriteLine("Enter a positive integer: ");
            num = Convert.ToInt32(Console.ReadLine());
        }
        public void prime()
        {
            int count = 0;
            for (int i = 1; i <= num; i++)
            {
                if (num % i == 0)
                {
                    count++;
                }
            }

            if (count == 2)
            {
                Console.WriteLine(num + " is a prime number.");
            }
            else
            {
                Console.WriteLine(num + " is not a prime number.");
            }
        }
    }
}
class program1

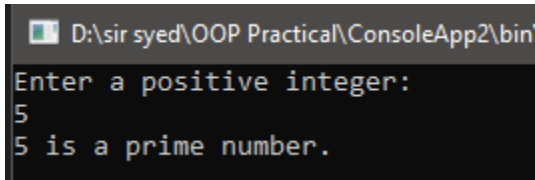
```

```

    {
        static void Main(string[] args) {
            Program obj = new Program();
            obj.input();
            obj.prime();
            Console.ReadLine();
        }
    }
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin
Enter a positive integer:
5
5 is a prime number.

```

Task 6: Write a program for four function calculators. (+, -, X, /) using switch -case

Code

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    internal class Program
    {
        public double num1, num2, result;
        public char a;
        public void input()
        {
            Console.WriteLine("Enter first number:");
            num1 = Convert.ToDouble(Console.ReadLine());

            Console.WriteLine("Enter second number:");
            num2 = Convert.ToDouble(Console.ReadLine());

            Console.WriteLine("Enter operator (+, -, *, /):");
            a = Convert.ToChar(Console.ReadLine());
        }

        public void add()
        {
            result = num1 + num2;
            Console.WriteLine("Result: " + result);
        }

        public void sub() {
            result = num1 - num2;
            Console.WriteLine("Result: " + result);
        }

        public void mul()
        {
            result = num1 * num2;
        }

        public void div()
        {
            result = num1 / num2;
        }
    }
}

class program1
{
    static void Main(string[] args) {
        Program obj = new Program();
        obj.input();
        switch (obj.a)

```

```

        {
            case '+':
                obj.add();
                break;

            case '-':
                obj.sub();
                break;

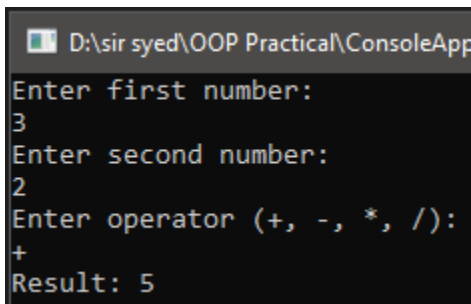
            case '*':
                obj.mul();
                break;

            case '/':
                obj.div();
                break;

            default:
                Console.WriteLine("Error: Invalid operator.");
                break;
        }
        Console.ReadLine();
    }
}
}

```

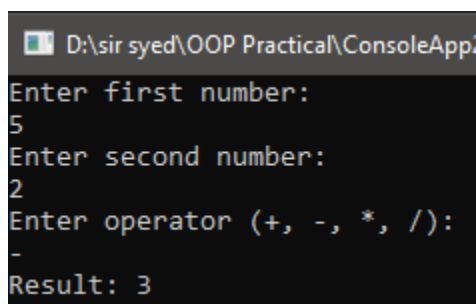
Output



```

D:\sir syed\OOP Practical\ConsoleApp2
Enter first number:
3
Enter second number:
2
Enter operator (+, -, *, /):
+
Result: 5

```



```

D:\sir syed\OOP Practical\ConsoleApp2
Enter first number:
5
Enter second number:
2
Enter operator (+, -, *, /):
-
Result: 3

```

Task 7: Print the list of numbers from 100 to 1. For each number print whether it is even or odd using do while loop.

Code

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    class Evenodd
    {
        public int num;
        public void input()
        {
            Console.WriteLine("Enter number");
            num = Convert.ToInt32(Console.ReadLine());
        }
        public void Decision()
        {
            int i = 100;

            do
            {
                if (i % 2 == 0)
                {
                    Console.WriteLine(i + " is even.");
                }
                else
                {

```

```

        Console.WriteLine(i + " is odd.");
    }
    i--;
} while (i >= 1);
}
}
class Program2
{
    static void Main(string[] args)
    {
        Evenodd obj = new Evenodd();
        obj.input();
        obj.Decision();
        Console.ReadLine();
    }
}
}

```

Output

```

D:\sir syed\OOP Practical\
100 is even.
99 is odd.
98 is even.
97 is odd.
96 is even.
95 is odd.
94 is even.
93 is odd.
92 is even.
91 is odd.
90 is even.
89 is odd.
88 is even.
87 is odd.
86 is even.
85 is odd.
84 is even.
83 is odd.
82 is even.
81 is odd.
80 is even.
79 is odd.
78 is even.
77 is odd.
76 is even.
75 is odd.
74 is even.
73 is odd.
72 is even.
71 is odd.
70 is even.
69 is odd.
68 is even.
67 is odd.
66 is even.
65 is odd.
64 is even.
63 is odd.
62 is even.
61 is odd.
60 is even.

```

```

D:\sir syed\OOP Practical\Cons
59 is odd.
58 is even.
57 is odd.
56 is even.
55 is odd.
54 is even.
53 is odd.
52 is even.
51 is odd.
50 is even.
49 is odd.
48 is even.
47 is odd.
46 is even.
45 is odd.
44 is even.
43 is odd.
42 is even.
41 is odd.
40 is even.
39 is odd.
38 is even.
37 is odd.
36 is even.
35 is odd.
34 is even.
33 is odd.
32 is even.
31 is odd.
30 is even.
29 is odd.
28 is even.
27 is odd.
26 is even.
25 is odd.
24 is even.
23 is odd.
22 is even.
21 is odd.
20 is even.
19 is odd.
18 is even.
17 is odd.
16 is even.

```

```

15 is odd.
14 is even.
13 is odd.
12 is even.
11 is odd.
10 is even.
9 is odd.
8 is even.
7 is odd.
6 is even.
5 is odd.
4 is even.
3 is odd.
2 is even.
1 is odd.

```


Task 8: Print the numbers that are divisible by 5 using do-while loop.

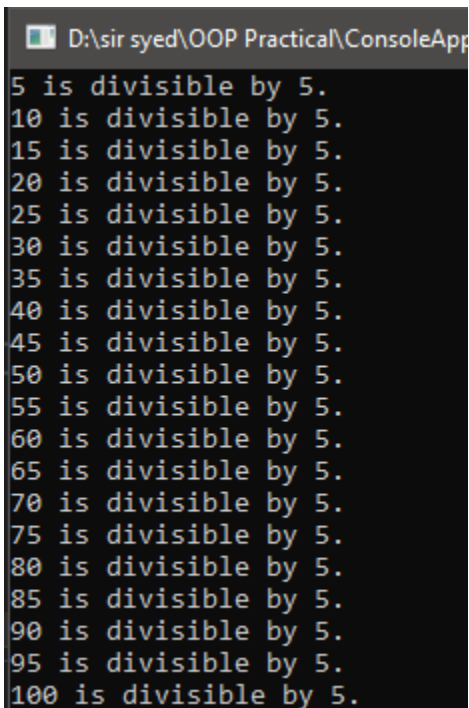
Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp2
{
    class Evenodd
    {
        public void Divisible()
        {
            int i = 1;

            do
            {
                if (i % 5 == 0)
                {
                    Console.WriteLine(i + " is divisible by 5.");
                }
                i++;
            } while (i <= 100);
        }
    }
    class Program2
    {
        static void Main(string[] args)
        {
            Evenodd obj = new Evenodd();
            obj.Divisible();
            Console.ReadLine();
        }
    }
}
```

Output



```
D:\sir syed\OOP Practical\ConsoleApp
5 is divisible by 5.
10 is divisible by 5.
15 is divisible by 5.
20 is divisible by 5.
25 is divisible by 5.
30 is divisible by 5.
35 is divisible by 5.
40 is divisible by 5.
45 is divisible by 5.
50 is divisible by 5.
55 is divisible by 5.
60 is divisible by 5.
65 is divisible by 5.
70 is divisible by 5.
75 is divisible by 5.
80 is divisible by 5.
85 is divisible by 5.
90 is divisible by 5.
95 is divisible by 5.
100 is divisible by 5.
```

Task 9: Write a program Season.java that takes two integers M and D and prints the season corresponding to month M (1 = January, 12 = December) and day D in the northern hemisphere.

Code

```
using System;

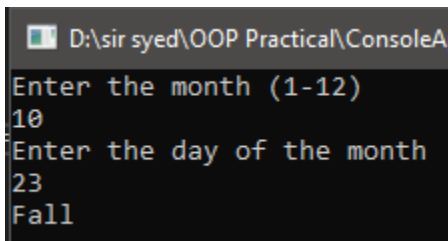
namespace ConsoleApp2
{
    class Program
    {
        public int M, D;
        public void input()
        {
            Console.WriteLine("Enter the month (1-12)");
            M = Convert.ToInt32(Console.ReadLine());

            Console.WriteLine("Enter the day of the month");
            D = Convert.ToInt32(Console.ReadLine());
        }

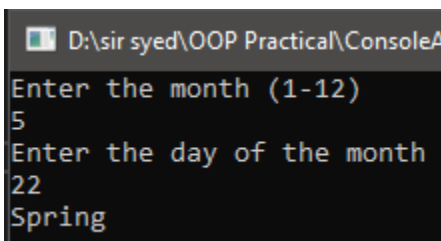
        public void season()
        {
            if (M == 1 || M == 2 || (M == 3 && D <= 20) || (M == 12 && D >= 21))
            {
                Console.WriteLine("Winter");
            }
            else if (M == 4 || M == 5 || (M == 3 && D >= 21) || (M == 6 && D < 21))
            {
                Console.WriteLine("Spring");
            }
            else if (M == 7 || M == 8 || (M == 6 && D >= 21) || (M == 9 && D < 22))
            {
                Console.WriteLine("Summer");
            }
            else
            {
                Console.WriteLine("Fall");
            }
        }
    }

    class Program2
    {
        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.input();
            obj.season();
            Console.ReadLine();
        }
    }
}
```

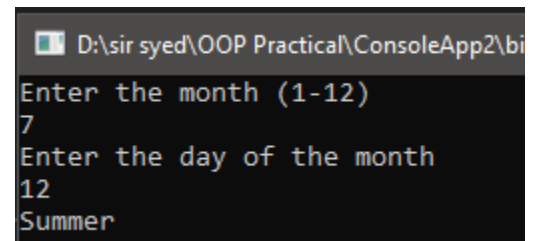
Output



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\net6.0\ConsoleApp2.exe
Enter the month (1-12)
10
Enter the day of the month
23
Fall
```



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\net6.0\ConsoleApp2.exe
Enter the month (1-12)
5
Enter the day of the month
22
Spring
```



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\net6.0\ConsoleApp2.exe
Enter the month (1-12)
7
Enter the day of the month
12
Summer
```

```
D:\sir syed\OOP Practical\ConsoleApp
Enter the month (1-12)
2
Enter the day of the month
12
Winter
```

Task 10: Write a program in Java to display the multiplication table of a given integer.

Code

using System;

```
namespace ConsoleApp2
{
    class Program
    {
        public int num1,result=0,i;
        public void input()
        {
            Console.WriteLine("Enter number");
            num1 = Convert.ToInt32(Console.ReadLine());
        }

        public void Multiplication()
        {
            Console.WriteLine("Multiplication table of {0} :", num1);
            for (i = 0; i <= 10; i++)
            {
                result = i * num1;
                Console.WriteLine("{0} * {1} : {2} ",num1,i,result);
            }
        }
    }
    class Program2
    {
        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.input();
            obj.Multiplication();
            Console.ReadLine();
        }
    }
}
```

Output

```
D:\sir syed\OOP Practical\ConsoleApp2\bin
Enter number
5
Multiplication table of 5 :
5 * 0 : 0
5 * 1 : 5
5 * 2 : 10
5 * 3 : 15
5 * 4 : 20
5 * 5 : 25
5 * 6 : 30
5 * 7 : 35
5 * 8 : 40
5 * 9 : 45
5 * 10 : 50
```

Lab # 4

Task 1: Write a program to explain get set properties.

Code

```
using System;

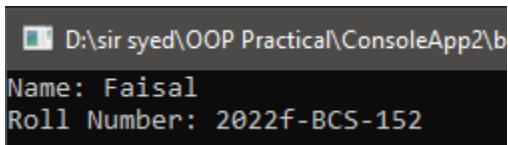
class Student
{
    private string name;
    private string rollNumber;

    public string Name
    {
        get { return name; }
        set { name = value; }
    }

    public string RollNumber
    {
        get { return rollNumber; }
        set { rollNumber = value; }
    }
}

class Program
{
    static void Main(string[] args)
    {
        Student student = new Student();
        student.Name = "Faisal";
        student.RollNumber = "2022f-BCS-152";
        Console.WriteLine("Name: " + student.Name);
        Console.WriteLine("Roll Number: " + student.RollNumber);
    }
}
```

Output



The screenshot shows a console window with the title bar "D:\sir syed\OOP Practical\ConsoleApp2\b". The output text is "Name: Faisal" followed by "Roll Number: 2022f-BCS-152" on the next line.

Task 2: Write a program to explain method in C#. Create a static function factorial() that accept a number from user and returns factorial of the number.

Code

```
using System;

class Program
{
    string name;
    string rollNumber;

    static int factorial(int n)
    {
        if (n == 0)
        {
            return 1;
        }
        else
        {
            return n * factorial(n - 1);
        }
    }

    static void Main(string[] args)
    {

```

```

        Program student = new Program();
        student.name = "Faisal";
        student.rollNumber = "2022f-BCS-152";
        Console.WriteLine("Name: " + student.name);
        Console.WriteLine("Roll Number: " + student.rollNumber);
        Console.WriteLine();

        Console.Write("Enter a number: ");
        int num = Convert.ToInt32(Console.ReadLine());

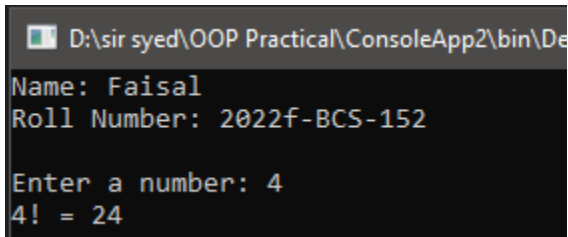
        int fact = factorial(num);

        Console.WriteLine(num + "! = " + fact);

        Console.ReadLine();
    }
}

```

Output



The screenshot shows a Windows command prompt window with the title "D:\sir syed\OOP Practical\ConsoleApp2\bin\De". The output of the program is as follows:

```

Name: Faisal
Roll Number: 2022f-BCS-152

Enter a number: 4
4! = 24

```

Task 3: Write a main method for your Vehicle class that creates a few vehicles and prints out their field values. Make the fields in your Vehicle class private, and add accessor and mutator methods for the fields

Code

```

using System;

class Vehicle
{
    private string make;
    private string model;
    private int year;

    public string GetMake()
    {
        return make;
    }

    public void SetMake(string make)
    {
        this.make = make;
    }

    public string GetModel()
    {
        return model;
    }

    public void SetModel(string model)
    {
        this.model = model;
    }

    public int GetYear()
    {
        return year;
    }

    public void SetYear(int year)
    {
        this.year = year;
    }
}

```

```

    }
}

class Program
{
    static void Main(string[] args)
    {
        Vehicle car1 = new Vehicle();
        Console.WriteLine("Name = Muhammad Faisal");
        Console.WriteLine("Roll number = 2022F-BCS-152");
        Console.WriteLine();

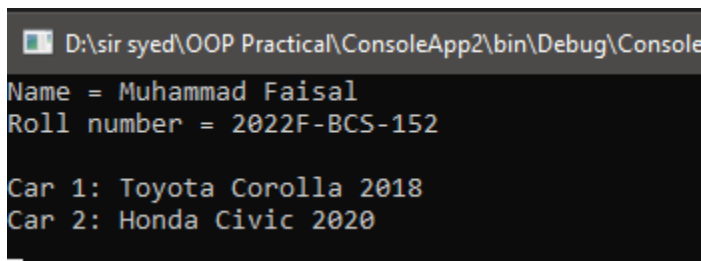
        car1.SetMake("Toyota");
        car1.SetModel("Corolla");
        car1.SetYear(2018);

        Vehicle car2 = new Vehicle();
        car2.SetMake("Honda");
        car2.SetModel("Civic");
        car2.SetYear(2020);

        Console.WriteLine("Car 1: " + car1.GetMake() + " " + car1.GetModel() + " " + car1.GetYear());
        Console.WriteLine("Car 2: " + car2.GetMake() + " " + car2.GetModel() + " " + car2.GetYear());
        Console.ReadLine();
    }
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Name = Muhammad Faisal
Roll number = 2022F-BCS-152

Car 1: Toyota Corolla 2018
Car 2: Honda Civic 2020

```

Task 4: Give the implementation of “PROTECTED INTERNAL” in C#.

Code

```

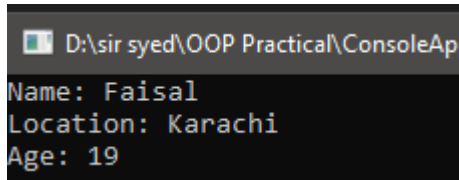
using System;

namespace ConsoleApp21
{
    class User
    {
        protected internal string Name;
        protected internal string Location;
        protected internal int Age;
        protected internal void GetUserDetails()
        {
            Console.WriteLine("Name: {0}", Name);
            Console.WriteLine("Location: {0}", Location);
            Console.WriteLine("Age: {0}", Age);
        }
    }

    class Program
    {
        static void Main(string[] args)
        {
            User obj = new User();
            obj.Name = "Faisal";
            obj.Location = "Karachi";
            obj.Age = 19;
            obj.GetUserDetails();
            Console.ReadLine();
        }
    }
}

```

Output



```
D:\sir syed\OOP Practical\ConsoleApp
Name: Faisal
Location: Karachi
Age: 19
```

Task 5: Write a program to show functionality of static class and static members. Create a static class which takes a string as input and checks if it's a palindrome or not.

Code

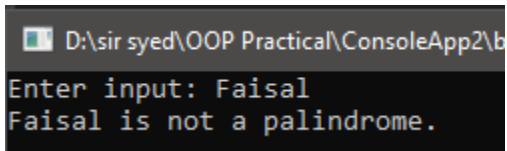
```
using System;

public static class StringHelper
{
    public static bool IsPalindrome(string str)
    {
        int length = str.Length;
        for (int i = 0; i < length / 2; i++)
        {
            if (str[i] != str[length - i - 1])
            {
                return false;
            }
        }
        return true;
    }
}

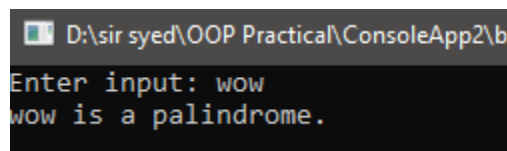
public class Program
{
    static void Main(string[] args)
    {
        string input = "Faisal";

        if (StringHelper.IsPalindrome(input))
        {
            Console.WriteLine($"{input} is a palindrome.");
        }
        else
        {
            Console.WriteLine($"{input} is not a palindrome.");
        }
        Console.ReadLine();
    }
}
```

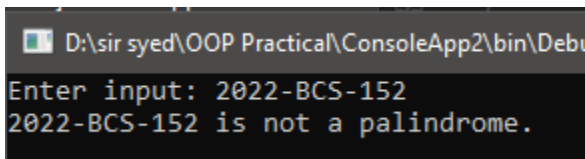
Output



```
D:\sir syed\OOP Practical\ConsoleApp2\b
Enter input: Faisal
Faisal is not a palindrome.
```



```
D:\sir syed\OOP Practical\ConsoleApp2\b
Enter input: wow
wow is a palindrome.
```



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debu
Enter input: 2022-BCS-152
2022-BCS-152 is not a palindrome.
```

Lab # 5

Task 1: Write a program which reads two arrays from the console and checks whether they are equal.

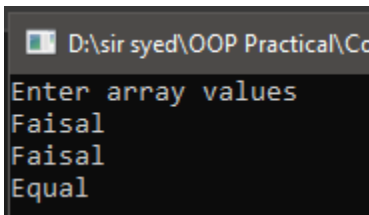
Code

```
using System;

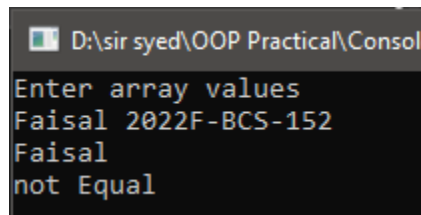
namespace ConsoleApp21
{
    class Program
    {
        string[] name = new string[3];
        public void array()
        {
            Console.WriteLine("Enter array values");
            for (int i = 0; i < 2; i++)
            {
                name[i] = Console.ReadLine();
            }

            if (name[0].Length == name[1].Length)
            {
                Console.WriteLine("Equal");
            }
            else
            {
                Console.WriteLine("not Equal");
            }
        }
        static void Main(string[] args)
        {
            Program obj = new Program();
            obj.array();
            Console.ReadLine();
        }
    }
}
```

Output



```
D:\sir syed\OOP Practical\ConsoleApp21>
Enter array values
Faisal
Faisal
Equal
```



```
D:\sir syed\OOP Practical\ConsoleApp21>
Enter array values
Faisal
2022F-BCS-152
not Equal
```

Task 2: Write a program which finds the most frequently occurring element in an array. Your array is {Rollno+2, 3*Rollno, Rollno, 35, 220, 50, 2*Rollno+4, 100}.

Code

```
using System;

namespace ConsoleApp21
{
    class Program
    {
        private int[] array = new int[] { 152 + 2, 3 * 152, 152, 35, 220, 50, 2 * 152 + 4, 100 };
        public void arr()
        {
            Array.Sort(array);

            int mostOccuring = array[0];
            int current = array[0];
            int count = 1;
            int maxCount = 1;
```



```

        for (int i = 1; i < array.Length; i++)
        {
            if (array[i] == current)
            {
                count++;
            }
            else
            {
                if (count > maxCount)
                {
                    maxCount = count;
                    mostOccuring = current;
                }
                current = array[i];
                count = 1;
            }
        }

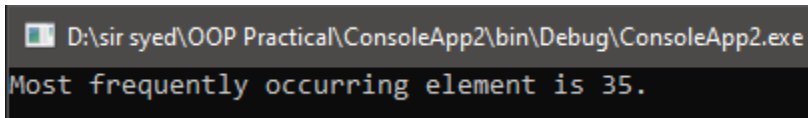
        if (count > maxCount)
        {
            maxCount = count;
            mostOccuring = current;
        }

        Console.WriteLine("Most frequently occurring element is {0}.", mostOccuring);
    }
}

class lab
{
    static void Main(string[] args)
    {
        Program obj = new Program();
        obj.arr();
        Console.ReadLine();
    }
}
}

```

Output



Task 3: Write a constructor program in which you make 3 constructors. One is for default constructors with default message ("THIS IS MY DATA OF CLASS(WRITE YOUR CLASS NAME)"), next is parameterized constructor which accept a string value i.e (your name and roll no in one string variable) and last one is also parameterized constructor which accept two numerical value i.e (AnyNumber+2, ROLLNO) and shows add of them. Initialize all constructors and shows output.

Code

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp21
{
    class Program
    {
        public string z;
        public int Rollnum;
        public int x, y, add;
        public Program()
        {
            Console.WriteLine("This is my data class (Section D)");
        }
    }
}

```

```

    public Program(string z)
    {
        this.z = z;
    }
    public Program(int x, int Rollnum)
    {
        this.x = x;
        this.Rollnum = Rollnum;
    }

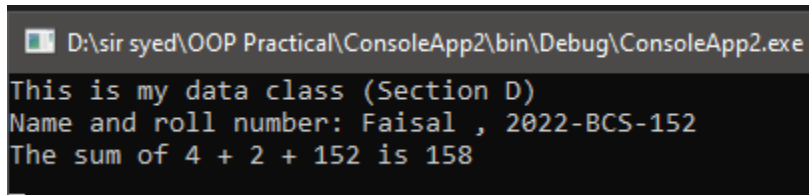
    public void sum()
    {
        add = x + 2 + this.Rollnum;
        Console.WriteLine("The sum of {0} + 2 + {1} is {2}", x, Rollnum, add);
    }

    public void print()
    {
        Console.WriteLine("Name and roll number: " + z);
    }
}
class lab
{
    static void Main(string[] args)
    {
        Program obj = new Program();
        Program obj1 = new Program("Faisal", 2022-BCS-152);
        Program obj2 = new Program(4, 152);

        obj1.print();
        obj2.sum();
        Console.ReadLine();
    }
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
This is my data class (Section D)
Name and roll number: Faisal , 2022-BCS-152
The sum of 4 + 2 + 152 is 158

```

Task 4: Write a program which has a class name “information” and manipulates bio data of bicycle (Name, location), car (Name of Driver, Car model and speed) and Lorry (Name, Model and Weight Carried with overloaded constructor. Note Use this operator as well.

Code

```

using System;

namespace ConsoleApp21
{
    class bicycle
    {
        string Name;
        string location;
        public bicycle(string Name, string location)
        {
            this.Name = Name;
            this.location = location;
        }
    }
}

```

```

    }
    public void print()
    {
        Console.WriteLine("Bio data of bicycle ");
        Console.WriteLine("Name of car: " + Name);
        Console.WriteLine("Location of car: " + location);
        Console.WriteLine();
    }
}

class car
{
    string Name;
    string Model;
    int speed;
    public car(string Name, string Model, int speed)
    {
        this.Name = Name;
        this.Model = Model;
        this.speed = speed;
    }
    public void print()
    {
        Console.WriteLine("Bio data of Car");
        Console.WriteLine("Name of Driver: " + Name);
        Console.WriteLine("Model of car: " + Model);
        Console.WriteLine("Speed of car: " + speed);
        Console.WriteLine();
    }
}

class Lorry
{
    string Name;
    string Model;
    double Weight;
    public Lorry(string Name, string Model, double Weight)
    {
        this.Name = Name;
        this.Model = Model;
        this.Weight = Weight;
    }
    public void print()
    {
        Console.WriteLine("Bio data of Lorry");
        Console.WriteLine("Name of Lorry: " + Name);
        Console.WriteLine("Model of Lorry: " + Model);
        Console.WriteLine("Weight of Lorry: " + Weight);
        Console.WriteLine();
    }
}

class information
{
    static void Main(string[] args)
    {
        bicycle obj = new bicycle("Hero", "Karachi");
        car obj1 = new car("Faisal", "Mustang", 150);
        Lorry obj2 = new Lorry("Tow truck", "Hyundai", 8600.7);
        obj.print();
        obj1.print();
        obj2.print();
        Console.ReadLine();
    }
}
}

```

Output

```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\net6.0\ConsoleApp2.exe
Bio data of bicycle
Name of car: Hero
Location of car: Karachi

Bio data of Car
Name of Driver: Faisal
Model of car: Mustang
Speed of car: 150

Bio data of Lorry
Name of Lorry: Tow truck
Model of Lorry: Hyundai
Weight of Lorry: 8600.7
```

Task 5: Give implementation of “Destructors” and also explain them.

In C#, destructor (finalizer) is used to destroy objects of class when the scope of an object ends. It has the same name as the class and starts with a tilde ~. For example,

Code

```
class Test
{
    ...
    //destructor
    ~Test()
    {
        ...
    }
}
```

Here, ~Test() is the destructor.

Code

```
using System;
namespace CsharpDestructor
{
    class Person
    {
        public Person()
        {
            Console.WriteLine("Constructor called.");
        }

        // destructor
        ~Person()
        {
            Console.WriteLine("Destructor called.");
        }

        public static void Main(string[] args)
        {
            //creates object of Person
            Person p1 = new Person();
        }
    }
}
```

Output

```
Constructor called.  
Destructor called.
```

In the above example, we have created a destructor ~person inside the person class. When we create an object of the person class, the constructor is called. After the scope of the object ends, object p1 is no longer needed. So, the destructor is called implicitly which destroys object p1.

Code

```
using System;  
namespace CsharpDestructor  
{  
    class Person  
    {  
        string name;  
        void getName()  
        {  
            Console.WriteLine("Name: " + name);  
        }  
  
        // destructor  
        ~Person()  
        {  
            Console.WriteLine("Destructor called.");  
        }  
  
        public static void Main(string[] args)  
        {  
  
            // creates object of Person  
            Person p1 = new Person();  
  
            p1.name = "Ed Sheeran";  
            p1.getName();  
        }  
    }  
}
```

Output

```
Name: Ed Sheeran  
Destructor called.
```

Lab # 6

Task 1:

Code

```
using System;
using System.Collections.Generic;
using System.Diagnostics;
using System.Linq;
using System.Runtime.ConstrainedExecution;
using System.Security.Principal;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp44
{
    class Account
    {
        public double account_balance;
        public double amount, debit_amount;

        public bool Method_Credit()
        {
            Console.WriteLine("Enter amount to credit");
            amount = Convert.ToDouble(Console.ReadLine());

            account_balance = account_balance + amount;

            Console.WriteLine("Current Balance after credit is " + account_balance);
            return true;
        }
        public bool Method_Debit()
        {
            Console.WriteLine("Enter amount to debit");
            debit_amount = Convert.ToDouble(Console.ReadLine());
            if(debit_amount > account_balance)
            {
                Console.WriteLine("Debit amount exceeded account balance.");
                return false;
            }
            else
            {
                account_balance = account_balance - debit_amount;
                Console.WriteLine("Current Balance after debit is " + account_balance);
                return true;
            }
        }

        public double getBalance()
        {
            return account_balance;
        }
    }
    class SavingsAccount : Account
    {
        public double interest_rate;

        public double CalculateInterest()
        {
            return interest_rate * account_balance;
        }
    }

    class CheckingAccount : Account
    {
        public double fee_per_transaction;

        public void Credit()
        {

```

```

        Method_Credit();
        if (true)
        {
            account_balance = account_balance - fee_per_transaction;
            Console.WriteLine("Current Balance after fee detuction is " + account_balance);
        }
    }

    public void Debit()
    {
        Method_Debit();
        if(true)
        {
            account_balance = account_balance - fee_per_transaction;
            Console.WriteLine("Current Balance after fee detuction is " + account_balance);
        }
    }
}

class Program
{
    static void Main(string[] args)
    {
        SavingsAccount obj = new SavingsAccount();
        CheckingAccount obj1 = new CheckingAccount();

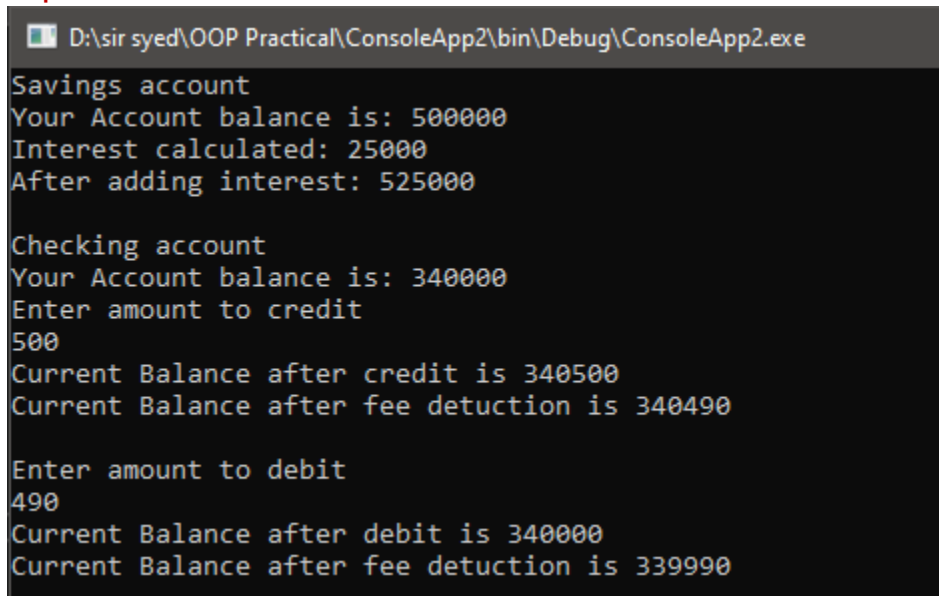
        Console.WriteLine("Savings account");
        obj.account_balance = 500000;
        obj.interest_rate = 0.05;
        Console.WriteLine("Your Account balance is: " + obj.getBalance());
        Console.WriteLine("Interest calculated: " + obj.CalculateInterest());
        Console.WriteLine("After adding interest: " +
(obj.CalculateInterest()+obj.account_balance));
        Console.WriteLine();

        Console.WriteLine("Checking account");
        obj1.account_balance = 340000;
        obj1.fee_per_transaction = 10;
        Console.WriteLine("Your Account balance is: " + obj1.getBalance());
        obj1.Credit();
        Console.WriteLine();
        obj1.Debit();

        Console.ReadLine();
    }
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Savings account
Your Account balance is: 500000
Interest calculated: 25000
After adding interest: 525000

Checking account
Your Account balance is: 340000
Enter amount to credit
500
Current Balance after credit is 340500
Current Balance after fee detuction is 340490

Enter amount to debit
490
Current Balance after debit is 340000
Current Balance after fee detuction is 339990

```

Task 2:

Code

```
using System;

namespace ConsoleApp44
{
    class Student
    {
        public string name, roll_no, section;
        public Student(string name, string roll_no, string section)
        {
            this.name = name;
            this.roll_no = roll_no;
            this.section = section;
        }
    }

    class Department : Student
    {
        public string department;
        public Department( string name, string roll_no, string section, string department) : base
(name, roll_no, section)
        {
            this.department = department;
        }
    }

    class Subjects : Department
    {
        public double average, per, total=0;
        public double[] s = new double[5];
        public Subjects(string name, string roll_no, string section, string department, double[] s) :
base (name, roll_no, section, department)
        {
            this.s = s;
        }

        public void calculate_percentage()
        {
            Console.WriteLine("Name: " + name);
            Console.WriteLine("Roll_no: " + roll_no);
            Console.WriteLine("Section: " + section);
            Console.WriteLine("Department: " + department);

            for (int i = 0; i < s.Length; i++)
            {
                total += s[i];
            }
            Console.WriteLine("Total Marks: "+total);

            average = (total / 5);
            per = ((total / 500) * 100);

            Console.WriteLine("Average of all subjects is: "+average);
            Console.WriteLine("Percentage is: "+per);
        }
    }

    class Program
    {
        static void Main(string[] args)
        {
            double[] marks1 = { 85, 90, 80, 80, 88 };
            Subjects obj = new Subjects("Bilal", "2022F-BCS-151", "D", "Computer Science", marks1);
            obj.calculate_percentage();
        }
    }
}
```



```

        Console.WriteLine();

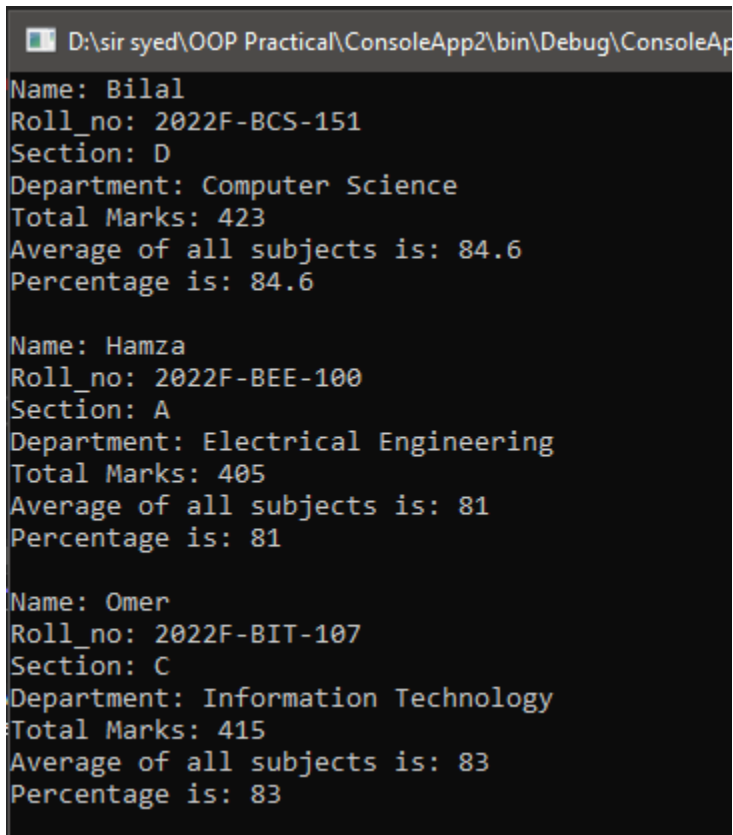
        double[] marks2 = { 75, 80, 90, 82, 78 };
        Subjects obj1 = new Subjects("Hamza", "2022F-BEE-100", "A", "Electrical
Engineering", marks2);
        obj1.calcualate_percentage();
        Console.WriteLine();

        double[] marks3 = { 89, 68, 89, 92, 77 };
        Subjects obj2 = new Subjects("Omer", "2022F-BIT-107", "C", "Information
Technology", marks3);
        obj2.calcualate_percentage();

        Console.ReadLine();
    }
}
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Name: Bilal
Roll_no: 2022F-BCS-151
Section: D
Department: Computer Science
Total Marks: 423
Average of all subjects is: 84.6
Percentage is: 84.6

Name: Hamza
Roll_no: 2022F-BEE-100
Section: A
Department: Electrical Engineering
Total Marks: 405
Average of all subjects is: 81
Percentage is: 81

Name: Omer
Roll_no: 2022F-BIT-107
Section: C
Department: Information Technology
Total Marks: 415
Average of all subjects is: 83
Percentage is: 83

```

Task 3:

Code

```

using System;
using System.Net;

namespace ConsoleApp21
{
    class Quadrilateral
    {
        private int x1, y1;
        private int x2, y2;
        private int x3, y3;
        private int x4, y4;
        public int X1
        {
            get { return x1; }
            set { x1 = value; }
        }
        public int X2

```

```

    {
        get { return x2; }
        set { x2 = value; }
    }
    public int X3
    {
        get { return x3; }
        set { x3 = value; }
    }
    public int X4
    {
        get { return x4; }
        set { x4 = value; }
    }
    public int Y1
    {
        get { return y1; }
        set { y1 = value; }
    }
    public int Y2
    {
        get { return y2; }
        set { y2 = value; }
    }
    public int Y3
    {
        get { return y3; }
        set { y3 = value; }
    }
    public int Y4
    {
        get { return y4; }
        set { y4 = value; }
    }
    public void Coordinates(int X1, int Y1, int X2, int Y2, int X3, int Y3, int X4, int Y4)
    {
        Console.WriteLine("x1 and y1 coordinate pair of the Quadrilateral is: " + "(" + X1 + " , " + Y1 + ")");
        Console.WriteLine("x2 and y2 coordinate pair of the Quadrilateral is: " + "(" + X2 + " , " + Y2 + ")");
        Console.WriteLine("x3 and y3 coordinate pair of the Quadrilateral is: " + "(" + X3 + " , " + Y3 + ")");
        Console.WriteLine("x4 and y4 coordinate pair of the Quadrilateral is: " + "(" + X4 + " , " + Y4 + ")");
        Console.WriteLine();
    }
}

class Trapezoid : Quadrilateral
{
    private int Base1, Base2, Height;

    public int BASE1
    {
        get { return Base1; }
        set { Base1 = value; }
    }
    public int HEIGHT
    {
        get { return Height; }
        set { Height = value; }
    }
    public int BASE2
    {
        get { return Base2; }
        set { Base2 = value; }
    }
}

```

```

        public void area1(int BASE1, int BASE2, int HEIGHT)
        {
            Console.WriteLine("Area of Trapezoid is: " + (BASE1 + BASE2 / 2) * (HEIGHT));
            Console.WriteLine();
        }
    }

    class Parallelogram : Trapezoid
    {
        private int Base, height1;
        public int BASE
        {
            get { return Base; }
            set { Base = value; }
        }
        public int HEIGHT1
        {
            get { return height1; }
            set { height1 = value; }
        }

        public void area2(int BASE, int HEIGHT)
        {
            Console.WriteLine("Area of Parallelogram is: " + (BASE) * (HEIGHT));
            Console.WriteLine();
        }
    }

    class Rectangle : Parallelogram
    {
        private int b2, h2;
        public int B2
        {
            get { return b2; }
            set { b2 = value; }
        }
        public int H2
        {
            get { return h2; }
            set { h2 = value; }
        }
        public void area3(int B2, int H2)
        {
            Console.WriteLine("Area of Rectangle is: " + (B2) * (H2));
            Console.WriteLine();
        }
    }

    class Square : Parallelogram
    {
        private int b3;
        public int B3
        {
            get { return b3; }
            set { b3 = value; }
        }
        public void area4(int B3)
        {
            Console.WriteLine("Area of Square is: " + (B3) * (B3));
            Console.WriteLine();
        }
    }

    class program
    {
        static void Main(string[] args)
        {
            Rectangle obj = new Rectangle();
            Square obj1 = new Square();
        }
    }

```

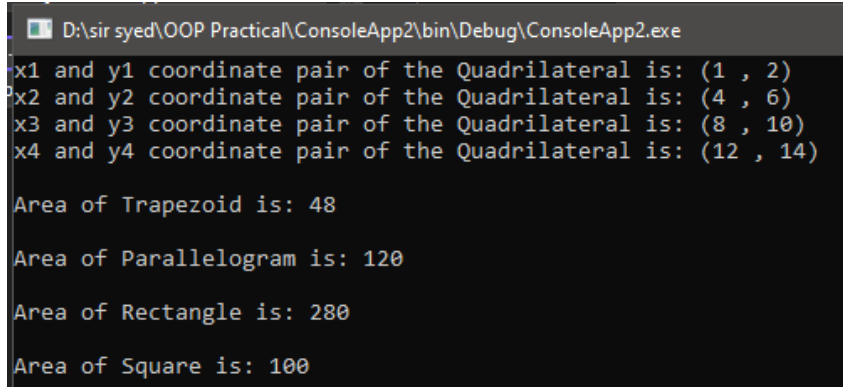
```

        obj.Coordinates(1, 2, 4, 6, 8, 10, 12, 14);
        obj.area1(12,24,2);
        obj.area2(10,12);
        obj.area3(14,20);
        obj1.area4(10);

        Console.ReadLine();
    }
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
x1 and y1 coordinate pair of the Quadrilateral is: (1 , 2)
x2 and y2 coordinate pair of the Quadrilateral is: (4 , 6)
x3 and y3 coordinate pair of the Quadrilateral is: (8 , 10)
x4 and y4 coordinate pair of the Quadrilateral is: (12 , 14)

Area of Trapezoid is: 48

Area of Parallelogram is: 120

Area of Rectangle is: 280

Area of Square is: 100

```

Task 4:

Code

```

using System;

namespace ConsoleApp44
{
    class Publication
    {
        public string title;
        public float price;

        public void getdata()
        {
            Console.Write("Enter title: ");
            title = Console.ReadLine();

            Console.Write("Enter price: ");
            price = Convert.ToInt64(Console.ReadLine());
        }

        public void putdata()
        {
            Console.WriteLine("The title of the book is: " + title);
            Console.WriteLine("The price of the book is: " + price);
        }
    }

    class Book : Publication
    {
        public int page_count;

        public void getdata()
        {
            base.getdata();
            Console.Write("Enter page count of the Book: ");
            page_count = Convert.ToInt32(Console.ReadLine());
        }

        public void putdata()
        {
            base.putdata();
            Console.Write("The page count of the Book is: " + page_count);
        }
    }
}

```

```

    }
}

class tape : Publication
{
    public float playing_time;

    public void getdata()
    {
        base.getdata();
        Console.WriteLine("Enter playing time of audiocassette (in minutes): " );
        playing_time = Convert.ToInt64(Console.ReadLine());
    }
    public void putdata()
    {
        base.putdata();
        Console.WriteLine("The playing time of audiocassette is: " + playing_time + " minutes");
    }
}

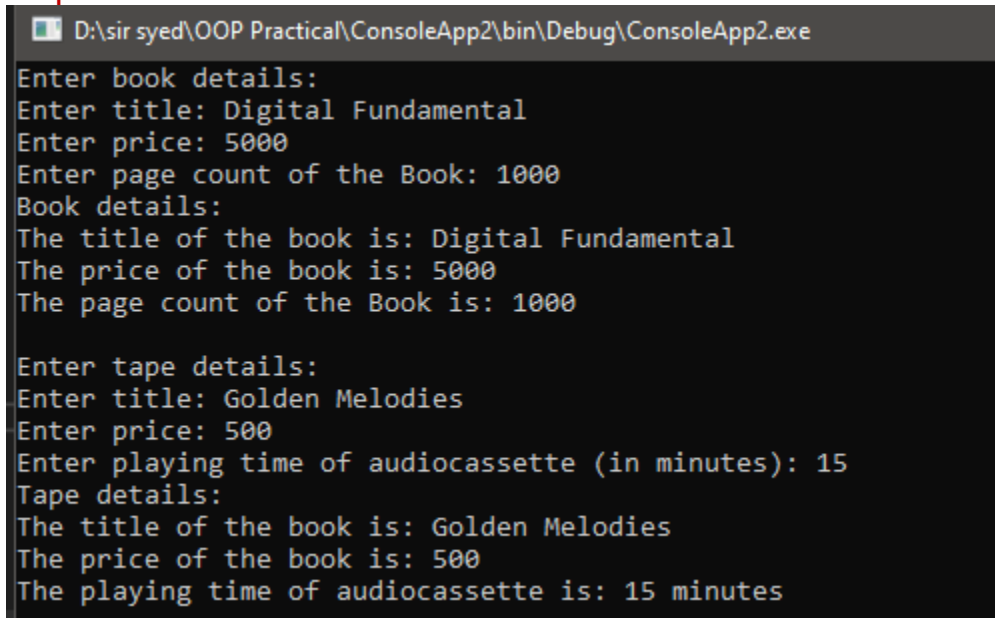
class Program
{
    static void Main(string[] args)
    {
        Book obj = new Book();
        Console.WriteLine("Enter book details:");
        obj.getdata();
        Console.WriteLine("Book details:");
        obj.putdata();
        Console.WriteLine();

        tape obj1 = new tape();
        Console.WriteLine("\nEnter tape details:");
        obj1.getdata();
        Console.WriteLine("Tape details:");
        obj1.putdata();

        Console.ReadLine();
    }
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Enter book details:
Enter title: Digital Fundamental
Enter price: 5000
Enter page count of the Book: 1000
Book details:
The title of the book is: Digital Fundamental
The price of the book is: 5000
The page count of the Book is: 1000

Enter tape details:
Enter title: Golden Melodies
Enter price: 500
Enter playing time of audiocassette (in minutes): 15
Tape details:
The title of the book is: Golden Melodies
The price of the book is: 500
The playing time of audiocassette is: 15 minutes

```

Task 5:

Code

```
using System;

namespace ConsoleApp44
{
    class Human
    {
        public string name, gender;

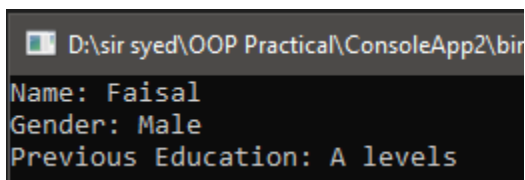
        public Human(string name, string gender)
        {
            this.name = name;
            this.gender = gender;
        }
    }
    class Student : Human
    {
        public string previous_education;

        public Student(string name, string gender, string previous_education): base(name,gender)
        {
            this.previous_education = previous_education;
        }

        public void Details()
        {
            Console.WriteLine("Name: " + name);
            Console.WriteLine("Gender: " + gender);
            Console.WriteLine("Previous Education: " + previous_education);
        }
    }

    class Program
    {
        static void Main(string[] args)
        {
            Student obj = new Student("Faisal", "Male", "A levels");
            obj.Details();
            Console.ReadLine();
        }
    }
}
```

Output



D:\sir syed\OOP Practical\ConsoleApp2\bin

Name: Faisal
Gender: Male
Previous Education: A levels

Lab # 7

Task 1:

Code

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp44
{
    class Vehicle
    {
        public string model, color, engine;
        public int door;
        public double stop_point, stop_time, start_point, start_time, total_time, total_distance,
speed;

        public Vehicle(string model, string color, string engine, int door)
        {
            this.model = model;
            this.color = color;
            this.engine = engine;
            this.door = door;
        }

        public virtual double calculate_speed()
        {
            return 0;
        }
    }
    class Car : Vehicle
    {
        public Car(string model, string color, string engine, int door) : base(model, color, engine,
door)
        {
            this.model = model;
            this.color = color;
            this.engine = engine;
            this.door = door;
        }
        public void start()
        {
            Console.WriteLine("Calculating speed of Car");
            Console.WriteLine();
            Console.WriteLine("Enter the starting point (in km)");
            start_point = Convert.ToDouble(Console.ReadLine());

            Console.WriteLine("Enter the starting time (in hours)");
            start_time = Convert.ToDouble(Console.ReadLine());
        }
        public void stop()
        {
            Console.WriteLine("Enter the stopping point (in km)");
            stop_point = Convert.ToDouble(Console.ReadLine());

            Console.WriteLine("Enter the stopping time (in hours)");
            stop_time = Convert.ToDouble(Console.ReadLine());
        }

        public void run()
        {
            total_distance = stop_point - start_point;
            total_time = stop_time - start_time;
        }
    }
}
```

```

    }
    public override double calculate_speed()
    {
        return total_distance / total_time;
    }
}

class Bike : Vehicle
{
    public Bike(string model, string color, string engine, int door) : base(model, color, engine,
door)
    {
        this.model = model;
        this.color = color;
        this.engine = engine;
        this.door = door;
    }
    public void start()
    {
        Console.WriteLine("Calculating speed of Bike");
        Console.WriteLine();
        Console.WriteLine("Enter the starting point (in km)");
        start_point = Convert.ToDouble(Console.ReadLine());

        Console.WriteLine("Enter the starting time (in hours)");
        start_time = Convert.ToDouble(Console.ReadLine());
    }
    public void stop()
    {
        Console.WriteLine("Enter the stopping point (in km)");
        stop_point = Convert.ToDouble(Console.ReadLine());

        Console.WriteLine("Enter the stopping time (in hours)");
        stop_time = Convert.ToDouble(Console.ReadLine());
    }

    public void run()
    {
        total_distance = stop_point - start_point;
        total_time = stop_time - start_time;
    }
    public override double calculate_speed()
    {
        return total_distance / total_time;
    }
}

class Program
{
    static void Main(string[] args)
    {
        Car obj = new Car("Civic", "Black", "1800CC", 4);
        Bike obj1 = new Bike("CD70", "Black", "70CC", 0);
        obj.start();
        obj.stop();
        obj.run();
        Console.WriteLine("Speed of car was " + obj.calculate_speed() + "km/h");
        Console.WriteLine();

        obj1.start();
        obj1.stop();
        obj1.run();
        Console.WriteLine("Speed of Bike was " + obj1.calculate_speed() + "km/h");
        Console.ReadLine();
    }
}

```



```
}
```

Output

```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Calculating speed of Car
Enter the starting point (in km)
0
Enter the starting time (in hours)
0
Enter the stopping point (in km)
150
Enter the stopping time (in hours)
3
Speed of car was 50km/h

Calculating speed of Bike
Enter the starting point (in km)
0
Enter the starting time (in hours)
0
Enter the stopping point (in km)
500
Enter the stopping time (in hours)
5
Speed of Bike was 100km/h
```

Task 2: Write a program to demonstrates sealed modifiers

Code

```
using System;

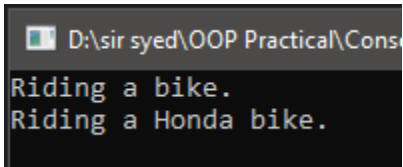
class Bike
{
    public virtual void bike()
    {
        Console.WriteLine("Riding a bike.");
    }
}

class Honda : Bike
{
    public sealed override void bike()
    {
        Console.WriteLine("Riding a Honda bike.");
    }
}

class Program
{
    static void Main()
    {
        Bike obj = new Bike();
        obj.bike();

        obj = new Honda();
        obj.bike();
        Console.ReadLine();
    }
}
```

Output



```
D:\sir syed\OOP Practical\Cons
Riding a bike.
Riding a Honda bike.
```

Task 3:

Code

```
using System;

namespace ConsoleApp44
{
    class Human
    {
        public int totaldistance = 0;
        public float total_distance = 0;
        public void Walk(int d)
        {
            totaldistance += d;
            Console.WriteLine("Walking: Distance covered: " + d);
            Console.WriteLine("Total distance: " + totaldistance);
        }

        public void Run(int d)
        {
            totaldistance += d;
            Console.WriteLine("Running: Distance covered: " + d);
            Console.WriteLine("Total distance: " + totaldistance);
        }

        public void Walk(float x)
        {
            total_distance += x;
            Console.WriteLine("Walking: Distance covered: " + x);
            Console.WriteLine("Total distance: " + total_distance);
        }

        public void Run(float x)
        {
            total_distance += x;
            Console.WriteLine("Running: Distance covered: " + x);
            Console.WriteLine("Total distance: " + total_distance);
        }

        public void Walk(string location)
        {
            Console.WriteLine("walking in: " +location);
        }

        public void Run(string location)
        {
            Console.WriteLine("running in: "+location);
        }
    }

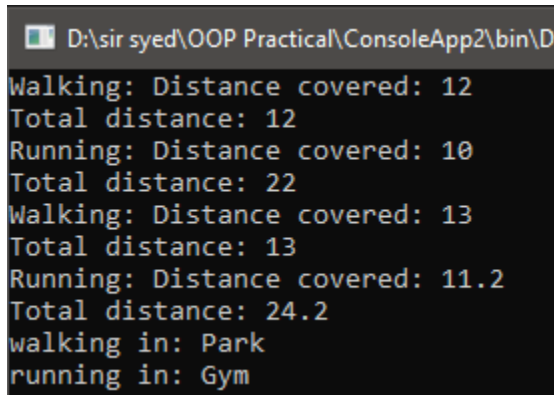
    class Program
    {
        static void Main(string[] args)
        {
            Human obj = new Human();
            obj.Walk(12);
            obj.Run(10);
            obj.Walk(13.0f);
            obj.Run(11.2f);
            obj.Walk("Park");
        }
    }
}
```

```

        obj.Run("Gym");
        Console.ReadLine();
    }
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug
Walking: Distance covered: 12
Total distance: 12
Running: Distance covered: 10
Total distance: 22
Walking: Distance covered: 13
Total distance: 13
Running: Distance covered: 11.2
Total distance: 24.2
walking in: Park
running in: Gym

```

Task 4: Implement the inheritance of Train and ReverseTrain classes having member function EngineWork taking two parameters Source and Destination with Data Types String. Train class should display Source as starting and Destination as end as output, whereas ReverseTrain should display Destination as starting and Source as end.

Code

```

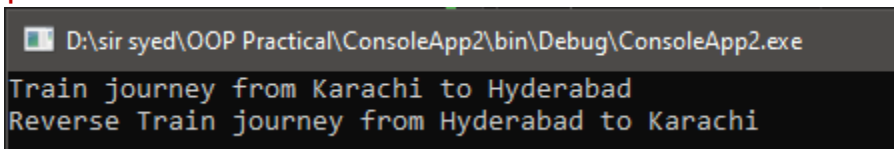
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ConsoleApp44
{
    class Train
    {
        public virtual void EngineWork(String Source, String Destination)
        {
            Console.WriteLine("Train journey from {0} to {1}", Source, Destination);
        }
    }
    class ReverseTrain: Train
    {
        public override void EngineWork(String Source, String Destination)
        {
            Console.WriteLine("Reverse Train journey from {0} to {1}", Destination, Source);
        }
    }
    class Program
    {
        static void Main(string[] args)
        {
            Train obj = new Train();
            obj.EngineWork("Karachi", "Hyderabad");
            obj = new ReverseTrain();
            obj.EngineWork("Karachi", "Hyderabad");

            Console.ReadLine();
        }
    }
}

```

Output



```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Train journey from Karachi to Hyderabad
Reverse Train journey from Hyderabad to Karachi

```

Task 5: Modify the member function of ReverseTrain of Exercise 3 such that it should also take extra parameter of JourneyTime and display in output concatenated at end of output.

Code

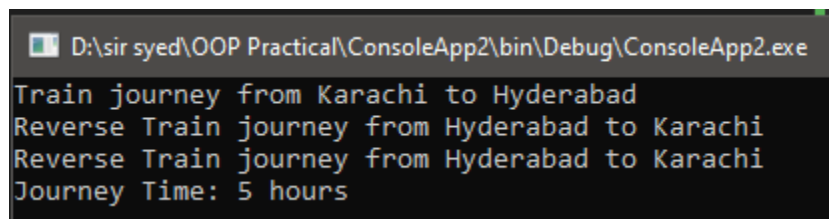
```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Xml.Serialization;

namespace ConsoleApp44
{
    class Train
    {
        public virtual void EngineWork(String Source, String Destination)
        {
            Console.WriteLine("Train journey from {0} to {1}", Source, Destination);
        }
    }
    class ReverseTrain : Train
    {
        public override void EngineWork(String Source, String Destination)
        {
            Console.WriteLine("Reverse Train journey from {0} to {1}", Destination, Source);
        }

        public void EngineWork(String Source, String Destination, int JourneyTime)
        {
            Console.WriteLine("Reverse Train journey from {0} to {1}", Destination, Source);
            Console.WriteLine("Journey Time: {0} hours", JourneyTime);
        }
    }
    class Program
    {
        static void Main(string[] args)
        {
            Train obj = new Train();
            obj.EngineWork("Karachi", "Hyderabad");
            ReverseTrain obj1 = new ReverseTrain();
            obj1.EngineWork("Karachi", "Hyderabad");
            obj1.EngineWork("Karachi", "Hyderabad", 5);

            Console.ReadLine();
        }
    }
}
```

Output



The screenshot shows a console window titled "D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe". The output text is as follows:

```
Train journey from Karachi to Hyderabad
Reverse Train journey from Hyderabad to Karachi
Reverse Train journey from Hyderabad to Karachi
Journey Time: 5 hours
```

Lab # 8

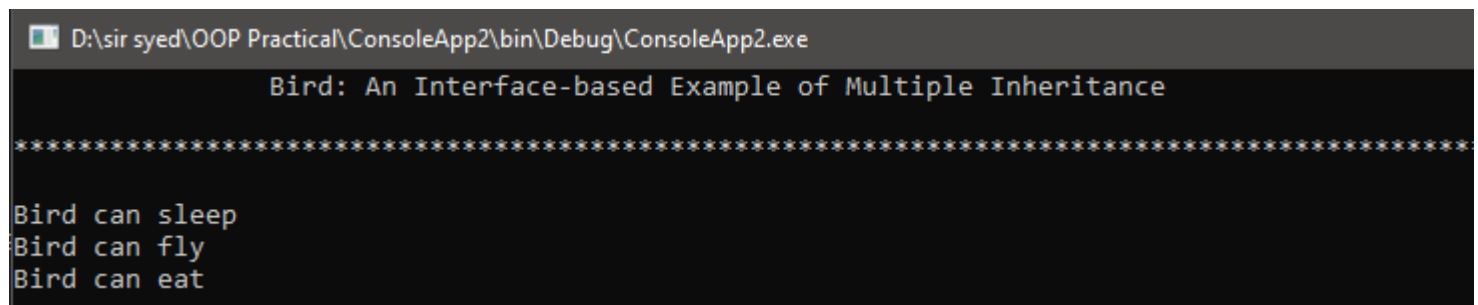
Task 1: Write a program to demonstrates the multiple inheritance

Code

```
using System;
using System.Diagnostics;
using System.Security.Cryptography.X509Certificates;
using System.Xml.Linq;

namespace ConsoleApp44
{
    public interface IAnimal
    {
        void sleep();
        void eat();
    }
    public interface IFly
    {
        void fly();
    }
    class Bird : IAnimal, IFly
    {
        public void sleep()
        {
            Console.WriteLine("Bird can sleep");
        }
        public void fly() {
            Console.WriteLine("Bird can fly");
        }
        public void eat()
        {
            Console.WriteLine("Bird can eat");
        }
    }
    class Program
    {
        static void Main(string[] args)
        {
            Bird obj = new Bird();
            Console.WriteLine("\t\tBird: An Interface-based Example of Multiple Inheritance");
            Console.WriteLine();
            for(int x = 0; x < 120; x++)
            {
                Console.Write("*");
            }
            Console.WriteLine();
            obj.sleep();
            obj.fly();
            obj.eat();
            Console.ReadLine();
        }
    }
}
```

Output



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe
Bird: An Interface-based Example of Multiple Inheritance
*****
Bird can sleep
Bird can fly
Bird can eat
```

Task 2: Write a program to demonstrates abstract modifiers

Code

```
using System;
using System.Diagnostics;
using System.Security.Cryptography.X509Certificates;
using System.Xml.Linq;

namespace ConsoleApp44
{
    public abstract class Employee
    {
        private string name, discription;

        public string getname()
        {
            return name;
        }
        public void setname(string name)
        {
            this.name = name;
        }
        public string getdiscription()
        {
            return discription;
        }
        public void setdiscription(string discription)
        {
            this.discription = discription;
        }

        public abstract void cal_salary();

        public void details()
        {
            Console.WriteLine("Employee name: "+name);
            Console.WriteLine("Employee job discription: " + discription);
        }
    }
    class Fulltime:Employee
    {
        private double salary;
        public double getsalary()
        {
            return salary;
        }
        public void setsalary(double salary)
        {
            this.salary = salary;
        }
        public override void cal_salary()
        {
            Console.WriteLine("Total salary is: "+ salary);
        }
    }
    class Parttime:Employee
    {
        private double HourlyRate;
        public int HoursWorked;
        public double getHourlyRate()
        {
            return HourlyRate;
        }
        public void setHourlyRate(double HourlyRate)
        {
            this.HourlyRate = HourlyRate;
        }
    }
}
```

```

        public override void cal_salary()
        {
            Console.WriteLine("Total salary is: " + (HourlyRate*HoursWorked));
        }
    }
}
class Program
{
    static void Main(string[] args)
    {
        Fulltime emp1 = new Fulltime();
        Parttime emp2 = new Parttime();

        Console.WriteLine("\n\t\tEmployee Salary Calculation with Abstraction in C#");
        Console.WriteLine();
        for(int x = 0; x < 120; x++)
        {
            Console.Write("*");
        }
        emp1.setname("Faisal");
        emp1.setdiscription("Developer");
        emp1.setsalary(2000000);
        emp2.setname("Areeb");
        emp2.setdiscription("Designer");
        emp2.setHourlyRate(10000);
        emp2.HoursWorked = 50;

        Console.WriteLine("\nFull time Employee Details");
        emp1.details();
        emp1.cal_salary();
        Console.WriteLine("\nPart time Employee Details");
        emp2.details();
        emp2.cal_salary();
        Console.ReadLine();
    }
}
}

```

Output

```

D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe

Employee Salary Calculation with Abstraction in C#
*****

Full time Employee Details
Employee name: Faisal
Employee job discription: Developer
Total salary is: 2000000

Part time Employee Details
Employee name: Areeb
Employee job discription: Designer
Total salary is: 500000

```

Task 3: Create an interface Itransaction contain void showTransaction(), double getAmount() ,Create class Transaction that inherit Itransaction and contains private string tCode; private string date; private double amount; add default constructor and parametric constructor to initialize values and implement void showTransaction(), double getAmount() in class.

Code

```
using System;
using System.Diagnostics;
using System.Security.Cryptography.X509Certificates;
using System.Xml.Linq;

namespace ConsoleApp44
{
    public interface Itransaction
    {
        void showTransaction();
        double getAmount();
    }

    class Transaction : Itransaction
    {
        private string tCode;
        private string date;
        private double amount;
        public double Amount
        {
            get { return amount; }
            set { amount = value; }
        }

        public string TCode
        {
            get { return tCode; }
            set { tCode = value; }
        }

        public string Date
        {
            get { return date; }
            set { date = value; }
        }

        public Transaction(){
        }
        public Transaction(string TCode, string Date, double Amount)
        {
            this.TCode = TCode;
            this.Date = Date;
            this.Amount = Amount;
        }
        public double getAmount()
        {
            return Amount;
        }
        public void showTransaction()
        {
            Console.WriteLine("\nTransaction Details");
            Console.WriteLine("Transaction Code: " + tCode);
            Console.WriteLine("Transaction date: " + date);
            Console.WriteLine("Transaction amount: " + getAmount());
        }
    }

    class Program
    {
        static void Main(string[] args)
        {
            Transaction obj = new Transaction("1001", "5/6/2023", 5000);
        }
    }
}
```



```

        Console.WriteLine("\n\t\tTransaction Management System with Interface and Class
Implementation in C#");
        Console.WriteLine();
        for(int x = 0; x < 120; x++)
        {
            Console.Write("*");
        }
        obj.showTransaction();

        Console.ReadLine();
    }
}
}

```

Output

```

Select D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe

Transaction Management System with Interface and Class Implementation in C#
*****
Transaction Details
Transaction Code: 1001
Transaction date: 5/6/2023
Transaction amount: 5000

```

Task 4: Create a C# program that implements an `IVehiculo` interface with two methods, one for `Drive` of type `void` and another for `Refuel` of type `bool` that has a parameter of type `integer` with the amount of gasoline to refuel. Then create a `Car` class with a builder that receives a parameter with the car's starting gasoline amount and implements the `Drive` and `Refuel` methods of the car. The `Drive` method will print on the screen that the car is Driving, if the gasoline is greater than 0. The `Refuel` method will increase the gasoline of the car and return `true`. To carry out the tests, create an object of type `Car` with 0 of gasoline in the Main of the program and ask the user for an amount of gasoline to refuel, finally execute the `Drive` method of the car.

Code

```

using System;
using System.Diagnostics;
using System.Security.Cryptography.X509Certificates;
using System.Xml.Linq;

namespace ConsoleApp44
{
    public interface IVehiculo
    {
        void Drive();
        bool Refuel(int amount_gas);
    }

    class Car : IVehiculo
    {
        public int starting_amount;

        public Car(int starting_amount)
        {
            this.starting_amount = starting_amount;
        }

        public bool Refuel(int amount_gas)
        {
            starting_amount = +amount_gas;
            return true;
        }

        public void Drive()
        {
            if (starting_amount > 0)
            {
                Console.WriteLine("The car is driving");
            }
        }
    }
}

```

```

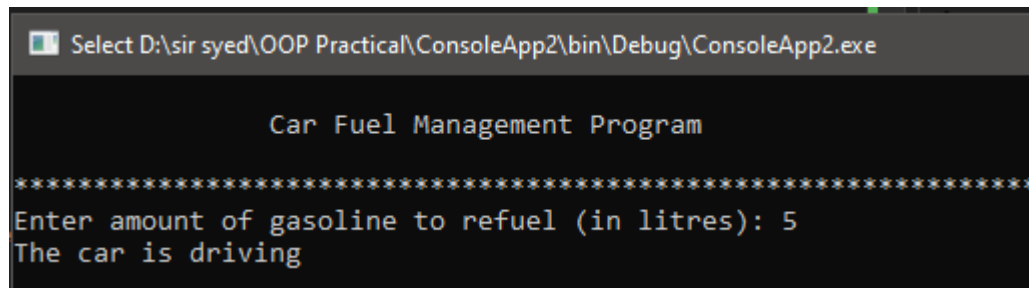
    }
    else
    {
        Console.WriteLine("The car is not driving");
    }
}
}
class Program
{
    static void Main(string[] args)
    {
        Car obj = new Car(0);

        Console.WriteLine("\n\t\tCar Fuel Management Program");
        Console.WriteLine();
        for(int x = 0; x < 120; x++)
        {
            Console.Write("*");
        }
        Console.Write("Enter amount of gasoline to refuel (in litres): ");
        int y = Convert.ToInt32(Console.ReadLine());
        obj.Refuel(y);
        obj.Drive();

        Console.ReadLine();
    }
}

```

Output



```

Select D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe

Car Fuel Management Program

*****
Enter amount of gasoline to refuel (in litres): 5
The car is driving

```

Lab # 9

Task 2: Write a program of throwing an exception when dividing by zero condition occurs?

Code

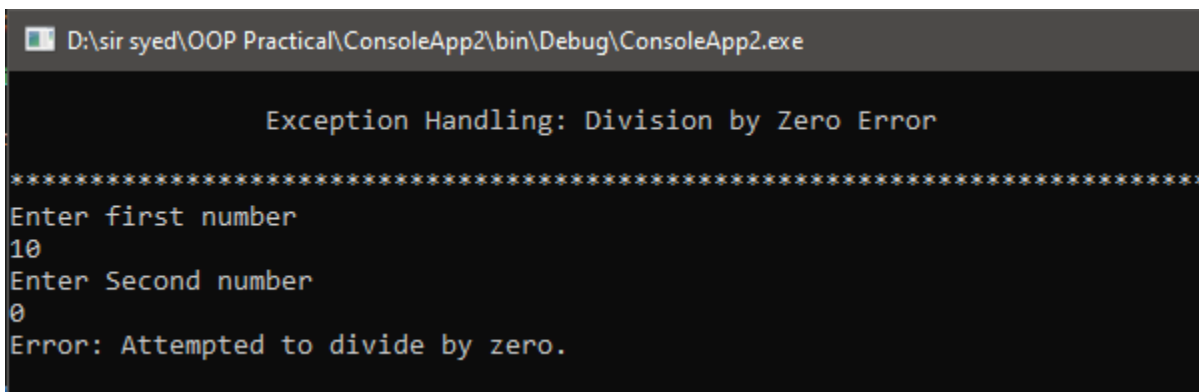
```
using System;

namespace consoleApp
{
    class A
    {
        public int num1,num2,result;

        public void Display()
        {
            Console.WriteLine("Enter first number");
            num1=Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("Enter Second number");
            num2 = Convert.ToInt32(Console.ReadLine());
            try
            {
                result = num1 / num2;
            }
            catch(DivideByZeroException ex)
            {
                Console.WriteLine("Error: " + ex.Message);
            }
        }
    }
    class Program
    {
        static void Main(string[] args)
        {
            A obj = new A();
            Console.WriteLine("\n\t\tException Handling: Division by Zero Error");
            Console.WriteLine();
            for (int x = 0; x < 120; x++)
            {
                Console.Write("*");
            }

            obj.Display();
            Console.ReadLine();
        }
    }
}
```

Output



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe

Exception Handling: Division by Zero Error
*****
Enter first number
10
Enter Second number
0
Error: Attempted to divide by zero.
```

Task 3: Write a program that if the user program wants to issue a book in library. The user program then can decide to generate a message and tell the user to check if the book is already issued etc?

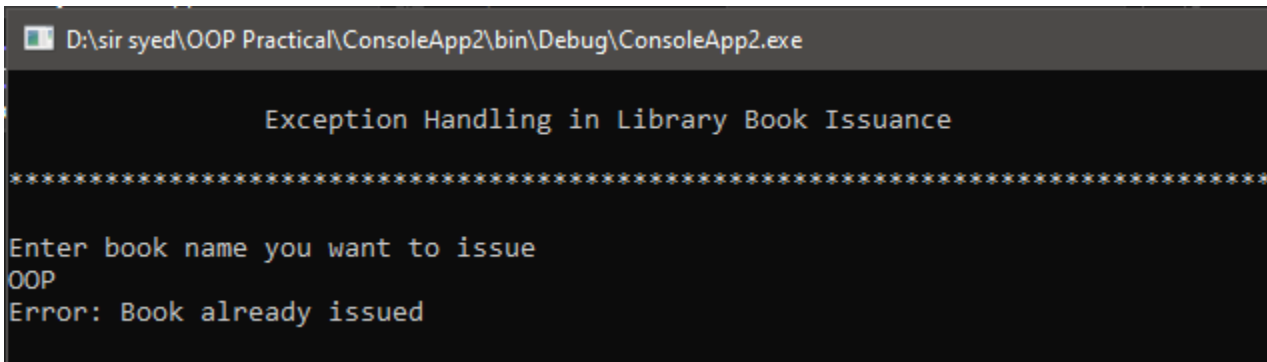
Code

```
using System;

namespace consoleApp
{
    class notvalid : Exception
    {
        public notvalid(string s) : base(s)
        {
        }
    }
    class A
    {
        public void Check(string y)
        {
            string[] issued = { "OOP", "DLD", "Calculus" };

            for (int i = 0; i < issued.Length; i++)
            {
                if (issued[i] == y)
                {
                    throw new notvalid("Book already issued");
                }
                else
                {
                    Console.WriteLine("Book available");
                }
            }
        }
    }
    class program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("\n\t\tException Handling in Library Book Issuance");
            Console.WriteLine();
            for (int x = 0; x < 120; x++)
            {
                Console.Write("*");
            }
            Console.WriteLine();
            try
            {
                A obj = new A();
                Console.WriteLine("Enter book name you want to issue");
                string x = Console.ReadLine();
                obj.Check(x);
            }
            catch (notvalid ex)
            {
                Console.WriteLine("Error: " + ex.Message);
            }
            Console.ReadLine();
        }
    }
}
```

Output



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe

Exception Handling in Library Book Issuance
*****

Enter book name you want to issue
OOP
Error: Book already issued
```

Task 4: Create an app that inputs miles driven and gallons used, and calculates miles per gallon. The example should use exception handling to process the `FormatExceptions` that occur when converting the input strings to doubles. If invalid data is entered, display a message informing the user.

Code

```
using consoleApp;
using System;

namespace consoleApp
{
    class A
    {
        public void Display()
        {
            try
            {
                Console.WriteLine("Miles Driven:");
                string milesInput = Console.ReadLine();
                double miles = Convert.ToDouble(milesInput);

                Console.WriteLine("Gallons Used:");
                string gallonsInput = Console.ReadLine();
                double gallons = Convert.ToDouble(gallonsInput);

                double milesPerGallon = miles / gallons;

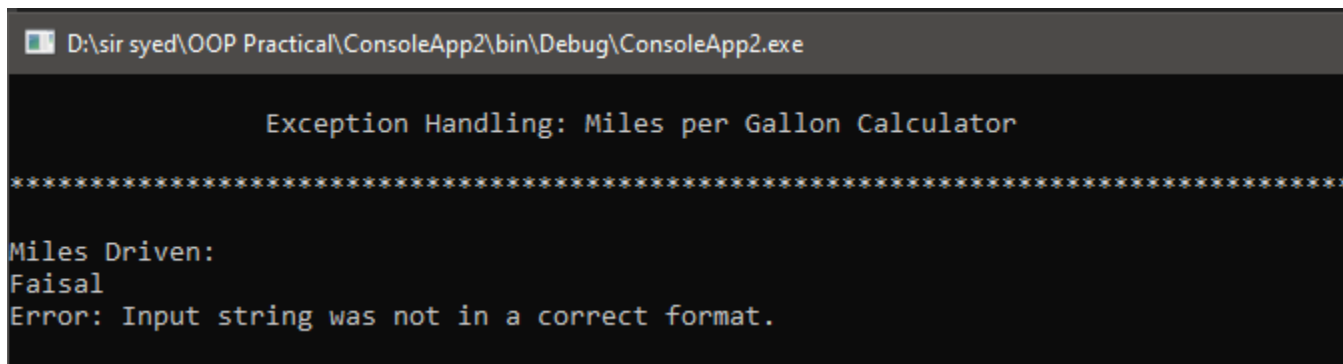
                Console.WriteLine("Miles per Gallon: " + milesPerGallon);
            }
            catch (FormatException ex)
            {
                Console.WriteLine("Error: " + ex.Message);
            }
        }
    }
}

class program
{
    static void Main(string[] args)
    {
        A obj = new A();
        Console.WriteLine("\n\t\tException Handling: Miles per Gallon Calculator");
        Console.WriteLine();
        for (int x = 0; x < 120; x++)
        {
            Console.Write("*");
        }
        Console.WriteLine();
        obj.Display();

        Console.ReadLine();
    }
}
```

```
}  
}
```

Output



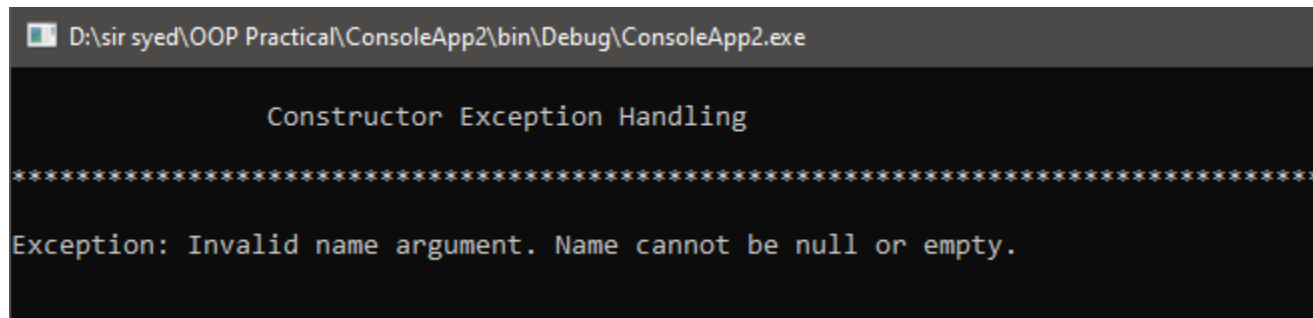
Task 5: Exceptions can be used to indicate problems that occur when an object is being constructed. Write a program that shows a constructor passing information about constructor failure to an exception handler. The exception thrown also should contain the arguments sent to the constructor

Code

```
using System;  
  
namespace ConstructorExceptionExample  
{  
    class MyClass  
    {  
        private string name;  
  
        public MyClass(string name)  
        {  
            if (string.IsNullOrEmpty(name))  
            {  
                throw new ArgumentException("Invalid name argument. Name cannot be null or empty.");  
            }  
  
            this.name = name;  
        }  
  
        public void Display()  
        {  
            Console.WriteLine("Name: " + name);  
        }  
    }  
  
    class Program  
    {  
        static void Main(string[] args)  
        {  
            Console.WriteLine("\n\t\tConstructor Exception Handling");  
            Console.WriteLine();  
            for (int x = 0; x < 120; x++)  
            {  
                Console.Write("*");  
            }  
            Console.WriteLine();  
  
            try  
            {  
                string invalidName = null;  
                MyClass obj = new MyClass(invalidName);  
                obj.Display();  
            }  
            catch (Exception ex)  
            {  
                Console.WriteLine("Exception: " + ex.Message);  
            }  
        }  
    }  
}
```

```
        Console.ReadLine();  
    }  
}
```

Output



```
D:\sir syed\OOP Practical\ConsoleApp2\bin\Debug\ConsoleApp2.exe  
  
Constructor Exception Handling  
*****  
Exception: Invalid name argument. Name cannot be null or empty.
```

Lab # 12

Task 1:



Code

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using System.Runtime.InteropServices;
using CustomControls.RJControls;
using Guna.UI.WinForms;

namespace WindowsFormsApp2
{
    public partial class Form1 : Form
    {
        [DllImport("Gdi32.dll", EntryPoint = "CreateRoundRectRgn")]
        private static extern IntPtr CreateRoundRectRgn
        (
            int nLeftRect,      // x-coordinate of upper-left corner
            int nTopRect,       // y-coordinate of upper-left corner
            int nRightRect,     // x-coordinate of lower-right corner
            int nBottomRect,    // y-coordinate of lower-right corner
            int nWidthEllipse,  // height of ellipse
            int nHeightEllipse // width of ellipse
        );

        public Form1()
        {
            InitializeComponent();
            this.FormBorderStyle = FormBorderStyle.None;
            Region = System.Drawing.Region.FromHrgn(CreateRoundRectRgn(0, 0, Width, Height, 20, 20));
        }

        private void rjButton1_Click(object sender, EventArgs e)
        {
            int num1 = int.Parse(gunaTextBox1.Text);
            int num2 = int.Parse(gunaTextBox2.Text);
            int result = num1 + num2;
            label5.Text = result.ToString();
        }
    }
}
```



```

private void rjButton2_Click(object sender, EventArgs e)
{
    int num1 = int.Parse(gunaTextBox1.Text);
    int num2 = int.Parse(gunaTextBox2.Text);
    int result = num1 - num2;
    label5.Text = result.ToString();
}

private void rjButton3_Click(object sender, EventArgs e)
{
    int num1 = int.Parse(gunaTextBox1.Text);
    int num2 = int.Parse(gunaTextBox2.Text);
    int result = num1 * num2;
    label5.Text = result.ToString();
}

private void rjButton4_Click(object sender, EventArgs e)
{
    int num1 = int.Parse(gunaTextBox1.Text);
    int num2 = int.Parse(gunaTextBox2.Text);
    int result = num1 / num2;
    label5.Text = result.ToString();
}

private void rjButton5_Click(object sender, EventArgs e)
{
    DialogResult result = MessageBox.Show("Are you sure you want to quit?", "OFF
application", MessageBoxButtons.YesNo);
    if (result == System.Windows.Forms.DialogResult.Yes)
    {
        Application.Exit();
    }
}

private void rjButton6_Click(object sender, EventArgs e)
{
    gunaTextBox1.Text = " ";
    gunaTextBox2.Text = " ";
    label5.Text = " ";
}
}
}

```

Output

Calculator

Number 1

Number 2

Result 5

Add

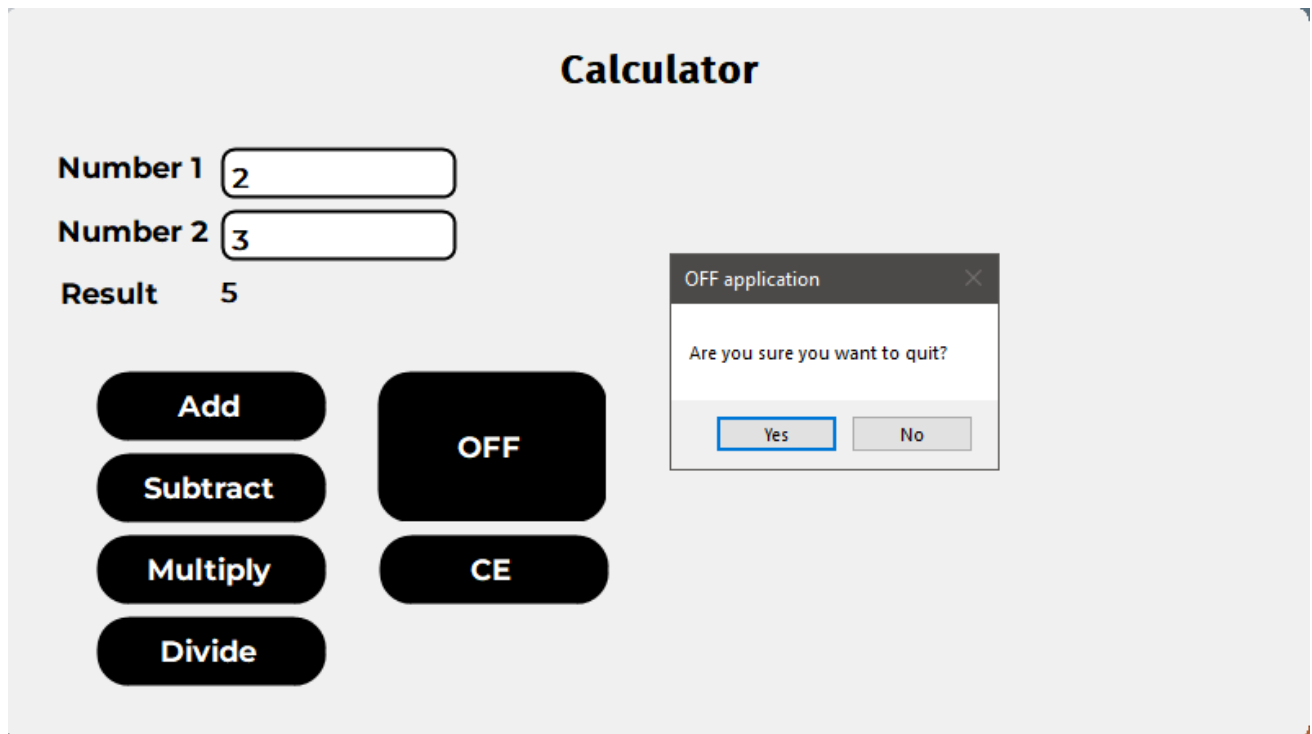
Subtract

Multiply

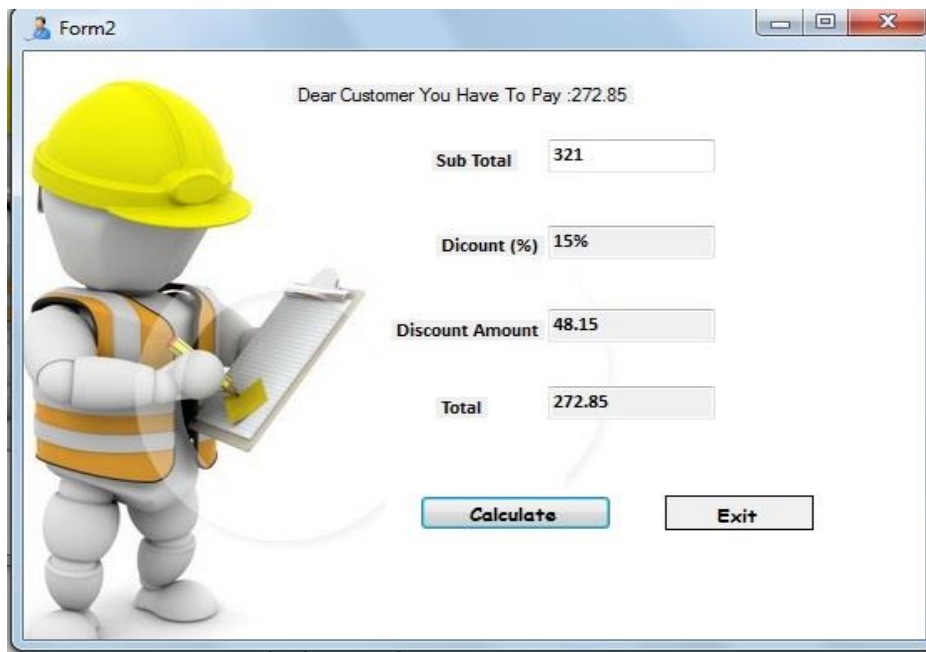
Divide

OFF

CE



Task 2:



Code

```
using System;
```

```
namespace WindowsFormsApp2
{
```

```
    public partial class Form1 : Form
    {
```

```
        [DllImport("Gdi32.dll", EntryPoint = "CreateRoundRectRgn")]
        private static extern IntPtr CreateRoundRectRgn
```

```
        (
            int nLeftRect,    // x-coordinate of upper-left corner
            int nTopRect,     // y-coordinate of upper-left corner
            int nRightRect,   // x-coordinate of lower-right corner
            int nBottomRect,  // y-coordinate of lower-right corner
            int nWidthEllipse, // height of ellipse
            int nHeightEllipse // width of ellipse
        );
```

```

public Form1()
{
    InitializeComponent();
    this.FormBorderStyle = FormBorderStyle.None;
    Region = System.Drawing.Region.FromHrgn(CreateRoundRectRgn(0, 0, Width, Height, 20, 20));
}

private void rjButton1_Click(object sender, EventArgs e)
{
    double Subtotal = Convert.ToDouble(gunaTextBox1.Text);
    double DiscountPercent = 0.0;
    if (Subtotal >= 500)
    {
        DiscountPercent = 0.2;
    }
    else if (Subtotal >= 250)
    {
        DiscountPercent = 0.15;
    }
    else if (Subtotal >= 100)
    {
        DiscountPercent = 0.1;
    }
    double DiscountAmount = Subtotal * DiscountPercent;
    double InvoiceAmount = Subtotal - DiscountAmount;
    gunaTextBox2.Text = DiscountPercent.ToString();
    gunaTextBox3.Text = DiscountAmount.ToString();
    gunaTextBox4.Text = InvoiceAmount.ToString();
    label8.Text = "Dear customer you have to pay: " + InvoiceAmount.ToString();
}

private void rjButton5_Click(object sender, EventArgs e)
{
    DialogResult result = MessageBox.Show("Are you sure you want to quit?", "Exit
application", MessageBoxButtons.YesNo);
    if (result == System.Windows.Forms.DialogResult.Yes)
    {
        Application.Exit();
    }
}
}
}

```

Output

Discount and Invoice Calculation



Sub Total	321
Discount (%)	0.15
Discount amount	48.15
Total	272.85

Dear customer you have to pay: 272.85

Discount and Invoice Calculation



Sub Total

321

Discount

Discount

Total

Exit application

Are you sure you want to quit?

Yes

No

Calculate

Exit

Dear customer you have to pay: 272.85

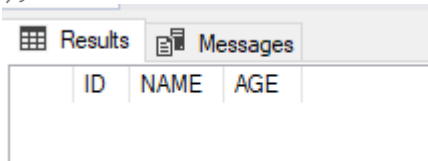
Lab # 13

Q1. Write the following SQL statements in Microsoft SQL Server

- Create Database Statement
- Create Table Statement
- Insert Into Statement
- Select Statement
- Update Table Statement
- SQL Drop Table Statement
- SQL Truncate Table Statement
- Alter Table Statement(Add/Drop Column)
- Truncate Table Statement

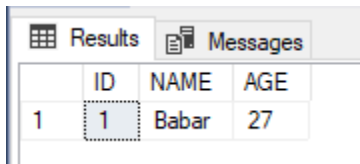
a) `CREATE DATABASE XYZ;`

b) `CREATE TABLE TABLE2(
ID INT PRIMARY KEY,
NAME VARCHAR(100) ,
AGE INT
);`



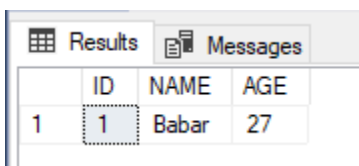
ID	NAME	AGE
----	------	-----

c) `INSERT INTO TABLE1 (NAME, AGE)
VALUES ('Babar', 27);`



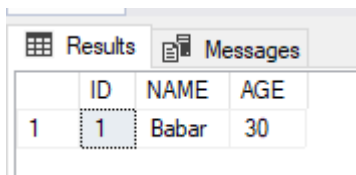
ID	NAME	AGE
1	Babar	27

d) `SELECT NAME, AGE
FROM TABLE1
WHERE AGE > 10;`



ID	NAME	AGE
1	Babar	27

e) `UPDATE TABLE2
SET AGE = 30
WHERE ID = 01;`

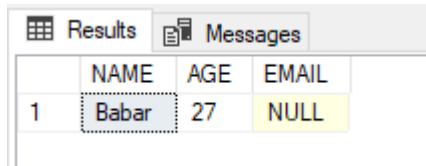


ID	NAME	AGE
1	Babar	30

f) `DROP TABLE TABLE2;`

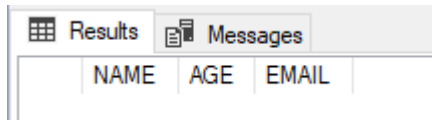
g) `TRUNCATE TABLE TABLE2;`

h) `ALTER TABLE TABLE1
ADD EMAIL VARCHAR(100);`



	NAME	AGE	EMAIL
1	Babar	27	NULL

i) `TRUNCATE TABLE TABLE1;`



	NAME	AGE	EMAIL
--	------	-----	-------

Q2. What is the main difference between OLEDB Provider and SqlClient?

The main difference between OLE DB Provider and SqlClient lies in their underlying technologies and the databases they are designed to work with:

1. OLE DB Provider:

OLE DB (Object Linking and Embedding, Database) is a Microsoft technology that provides a data access interface for accessing various types of data sources, including relational databases, spreadsheets, and text files. OLE DB is a component-based API that allows applications to access data using a set of COM (Component Object Model) interfaces.

The OLE DB Provider for SQL Server allows applications to access SQL Server databases using the OLE DB interface. It provides a consistent way to interact with SQL Server and other OLE DB-compliant data sources. OLE DB is a general-purpose data access technology that supports various databases and data formats.

2. SqlClient:

SqlClient is a data provider specifically designed for accessing Microsoft SQL Server databases. It is part of the .NET Framework's System.Data.SqlClient namespace and provides a streamlined, optimized, and efficient means of interacting with SQL Server.

SqlClient is built on top of the ADO.NET (ActiveX Data Objects for .NET) framework and is optimized for accessing SQL Server features and functionalities. It offers features like connection pooling, parameterized queries, support for SQL Server-specific data types, and other performance optimizations for SQL Server.

In summary, while both OLE DB Provider and SqlClient can be used to access SQL Server databases, OLE DB is a general-purpose data access technology that can work with various data sources, whereas SqlClient is a specialized data provider specifically designed for interacting with Microsoft SQL Server databases, offering better performance and SQL Server-specific optimizations.

Q3. Create an application that stores and retrieves/search/read data from database using ADO.NET in windows form C#.

```
using System;
using System.Configuration;
using System.Data;
using System.Data.SqlClient;
using System.Windows.Forms;

namespace WindowsFormsApp9
{
    public partial class Form1 : Form
    {
        public string connectionString;

        public Form1()
        {
            InitializeComponent();
            connectionString = ConfigurationManager.ConnectionStrings["xyy"].ConnectionString;
        }

        private void btnSave_Click(object sender, EventArgs e)
        {
        }

        private void btnRetrieve_Click(object sender, EventArgs e)
        {
        }

        private void btnSearch_Click(object sender, EventArgs e)
        {
            string searchKeyword = txtSearch.Text;

            using (SqlConnection con = new SqlConnection(connectionString))
            {
                string query = "SELECT Name, Email FROM Table4 WHERE Name LIKE @SearchKeyword";
                SqlCommand command = new SqlCommand(query, con);
                command.Parameters.AddWithValue("@SearchKeyword", "%" + searchKeyword + "%");

                con.Open();

                SqlDataReader reader = command.ExecuteReader();

                if (reader.HasRows)
                {
                    DataTable dataTable = new DataTable();
                    dataTable.Load(reader);
                    dataGridView1.DataSource = dataTable;
                }

                reader.Close();
            }
        }

        private void btnSave_Click_1(object sender, EventArgs e)
        {
            string name = txtName.Text;
            string email = txtEmail.Text;

            using (SqlConnection con = new SqlConnection(connectionString))
            {
                string query = "INSERT INTO TABLE4 (Name, Email) VALUES (@Name, @Email)";
                SqlCommand command = new SqlCommand(query, con);
                command.Parameters.AddWithValue("@Name", name);
                command.Parameters.AddWithValue("@Email", email);

                con.Open();
                int rowsAffected = command.ExecuteNonQuery();
                if (rowsAffected > 0)
                {
                }
            }
        }
    }
}
```


Lab # 14

Task

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Configuration;
using System.Data;
using System.Data.SqlClient;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace WindowsFormsApp9
{
    public partial class RegistrationForm : Form
    {
        public string connectionString;

        public RegistrationForm()
        {
            InitializeComponent();
            connectionString = ConfigurationManager.ConnectionStrings["xyy"].ConnectionString;
        }

        private void btnLogin_Click(object sender, EventArgs e)
        {
            string username = txtUsername.Text;
            string password = txtPassword.Text;
            string fathename = gunaTextBox2.Text;
            string item1 = checkBox1.Checked ? checkBox1.Text : "";
            string item2 = checkBox2.Checked ? checkBox2.Text : "";
            string item3 = checkBox3.Checked ? checkBox3.Text : "";
            string item4 = checkBox4.Checked ? checkBox4.Text : "";
            string item5 = checkBox5.Checked ? checkBox5.Text : "";

            using (SqlConnection connection = new SqlConnection(connectionString))
            {
                try
                {
                    connection.Open();

                    string query = "INSERT INTO Users (Username, Pass,Fathename,Education,University
) VALUES (@Username, @Pass, @Fathename,@Education,@University)";
                    SqlCommand command = new SqlCommand(query, connection);
                    command.Parameters.AddWithValue("@Username", username);
                    command.Parameters.AddWithValue("@Pass", password);
                    command.Parameters.AddWithValue("@Fathename", fathename);
                    if (!string.IsNullOrEmpty(item1))
                        command.Parameters.AddWithValue("@Education", item1);

                    if (!string.IsNullOrEmpty(item2))
                        command.Parameters.AddWithValue("@Education", item2);

                    if (!string.IsNullOrEmpty(item3))
                        command.Parameters.AddWithValue("@University", item3);
                    if (!string.IsNullOrEmpty(item4))
                        command.Parameters.AddWithValue("@University", item4);
                    if (!string.IsNullOrEmpty(item5))
                        command.Parameters.AddWithValue("@University", item5);

                    int rowsAffected = command.ExecuteNonQuery();
                }
            }
        }
    }
}
```

```

        if (rowsAffected > 0)
        {
            // Registration successful
            MessageBox.Show("Registration successful.");
            LoginForm obj = new LoginForm();
            obj.Show();
            obj.BringToFront();
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("An error occurred: " + ex.Message);
    }
}

}
}

using System;
using System.Configuration;
using System.Data;
using System.Data.SqlClient;
using System.Windows.Forms;
using static System.Windows.Forms.VisualStyles.VisualStyleElement.StartPanel;

namespace WindowsFormsApp9
{
    public partial class LoginForm : Form
    {
        public string connectionString;

        public LoginForm()
        {
            InitializeComponent();
            connectionString = ConfigurationManager.ConnectionStrings["xyy"].ConnectionString;
        }

        private void btnSave_Click_1(object sender, EventArgs e)
        {
            string username = txtUsername.Text;
            string password = txtPassword.Text;

            using (SqlConnection connection = new SqlConnection(connectionString))
            {
                try
                {
                    connection.Open();

                    string query = "SELECT COUNT(*) FROM Users WHERE Username = @Username AND Pass = @Pass";

                    SqlCommand command = new SqlCommand(query, connection);
                    command.Parameters.AddWithValue("@Username", username);
                    command.Parameters.AddWithValue("@Pass", password);

                    int result = Convert.ToInt32(command.ExecuteScalar());

                    if (result > 0)
                    {
                        // Successful login
                        MessageBox.Show("Login successful.");
                    }
                    else
                    {
                        // Invalid credentials
                        MessageBox.Show("Invalid username or password.");
                    }
                }
                catch (Exception ex)

```

```

        {
            MessageBox.Show("An error occurred: " + ex.Message);
        }
    }

    private void gunaAdvenceButton1_Click(object sender, EventArgs e)
    {
        RegistrationForm obj = new RegistrationForm();
        obj.Show();
        obj.BringToFront();
    }
}

```

The screenshot shows a Windows application window titled "RegistrationForm". The window has a light gray background and standard Windows window controls (minimize, maximize, close) in the top right corner. The form contains the following elements:

- Username:** A text input field containing "ABCD".
- Password:** A text input field containing "ABCD".
- Father Name:** A text input field containing "ABCD".
- Education:** Two checkboxes. "BS" is checked, and "Intermediate" is unchecked.
- Selected University:** Three checkboxes. "Ned" is unchecked, "SSUET" is checked, and "Iqra" is unchecked.
- Register:** A black button with white text located at the bottom center of the form.

A small dialog box is open on the right side of the window. It has a dark gray title bar with a close button (X). The dialog box contains the text "Registration successful." and an "OK" button at the bottom.

Username

ABCD

Register

Password

ABCD

Login

×

Login successful.

OK